

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EEE 211 — Past Questions & Materials
by Renance

Course: EEE 211
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 200

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Paper 1 of 5 — EEE 211 {2018/ 2019}

Level 200 • First Semester • PastQuestion

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Mh gent EXAMINATION: Firs Semeste

COURSE CODE: EEE 211

yy TIME ALLOWED: 2 Hrs

: SESSION: 2018/2019

COURSE TITLE: Basic Electrical Engineering =

DEGREE: B.ENG

INSTRUCTION: Answer FOUR Questions from both sections

= —————

SECTION A: Answer TWO Questions from this section

1. (a) State Maximum Power Transfer theorem, and justify the need for such a theorem, (4% marks)

(6) (i) State Superposition theorem, Explain why this theorem does not apply to power as an electrical

quantity, and state how power can be computed using this theorem. (5 marks)

(ii) Using Superposition theorem, calculate the current flowing through the 4 2 resistor in figure Q1 (b).

Also, compute the voltage drop and power dissipated in the 4 Q resistor, (8 marks)

52

ave ove

42

22 32

Figure QI =

e : f a maf ene

2. (a) Explain the following terms: @ Node (ii) via (iii) Branch (iv) Transient response. (6 marks)

(b) For the circuit of figure Q2 (b), (i) Write out the differential equation (ii) Compute the time-constant,

natural response and forced response (33 marks)

(c) Calculate the mesh currents and branch currents in the circuit of figure Q2 (c). (8 marks)

WA— MW

t=0 80 30

4Q

6a M2 Ma au RV

60V-T O1H }i(t=0)=0A =F ov

t WW a

Figure Q2 (b) Figure Q2(c) 5%

f

3. (a) State Kirchhoff's current law and Kirchhoff's voltage law, (3 marks)

(b) State Norton's theorem. (2% marks)

(c) (i) Using Thevenin theorem, calculate the current through the 8 Ω resistor in figure Q3 (c). (9 marks)

(ii) Draw the Norton equivalent circuit of the circuit obtained in Question 3c (i) and verify the value of the

current flowing through the 8 Ω resistor. (3 marks)

4

5a

| 12Vv 30

3Q 30

Figure Q3 (c)

SECTION B: Answer TWO Questions from this section

4. (a) State the laws of magnetism. (3 marks)

(b) Define the following: (i) magnetic field (ii) magnetic flux (iii) relative permeability (6 marks)

H

(c) An inductive coil of resistance 10 Ω and inductance 0.1 H is connected in parallel with a 150 μF capacitor |

to a variable frequency, 200 V supply. Find the resonant frequency at which the total current taken from |

the supply is in phase with the supply voltage. Find also the value of this current. Draw the circuit arrangement and the phasor diagrams. (8 marks)

5. (a) With the aid of a circuit diagram and all necessary equations, show that when an alternating voltage

is applied to a purely capacitive circuit, the resulting current leads the applied voltage by $\frac{\pi}{4}$ radians.

(8 marks)

(b) A magnetic circuit is excited by three coils as shown in figure Q5 (b). Calculate the flux produced in the

air gap. The material used for core is iron having relative permeability of 800, The length of the magnetic

'circuit-is 100 cm-with. an-air gap-of 2-cm in it/ The core has uniform cross-section of 6 cm^2 . Take $\mu_r = 42107$. (9% marks) .

sos

Figure Q5 (b)

6. (a) Enumerate the importance of RMS value in the field of Engineering. A

(b) The waveform of the voltage and current of a circuit are given by:

$v = 120 \sin(314t)$ and $i = 10 \sin(314t + \frac{\pi}{4})$. Calculate the values of the re:

are connected in series to form the circuit. Al

Calculate power consumed by the circuit,

(6 marks)

sistance and capacitance which

iso draw waveforms for current, voltage and phasor diagarar,

(115 marks)

aa

Li

Paper 2 of 5 — EEE 211 (2015/2016)

Level 200 • First Semester • PastQuestion

Z

FEDERAL pore OF PETROLEUM RESOURCES, EFFURUN
LEGE OF TECHNOLOGY

EXAMINATION: First Semester SESSION: 2015/2016

COURSE CODE: EEE 211

{ i; COURSE TITLE: Basic Electrical Enginering J

4 TIME ALLOWED: 2Hrs 30mins qi ergs), DEGREE: -B-ENG— — — — —_— -

ca

INSTRUCTION: Answer FOUR, Ghestinx with AT LEAST ONE Goes from each section

_SECTIONA___ ms |

1. (a) (i) State Coulomb's law ofelecttostatics. | - 2% (1 4 marks)

(ii) State the inverse square law. i : : (1 % marks)

(iii) Calculate: the electrostatic force of repulsitn in free space when 10 pm is the distance apart between two particles. Both

particles have en equal charge of 4×10^{-10} ? Cand ma of 6.68×10^{-31} Kg. (2 marks)

(b) (i) Define Magnetic Flux density.

-b4Gi) With the aid of simple diagrams, outline three similarities and three dissimilarities between a magnet circuit and an electtic

circuit. { H g GK marks)

(iii) A solenoid of length 20 cm has 500 turns and SA current flowing through it, Calculate the magnetizing force and_flux

density. wy ; (1 mark)

| 4

(c) (i) Explain briefly what you know about Inductors, (2 marks)

(ii) Mention two applications of an inductor. (i mark)

(ui) A circular magnetic core with relative per meability of 2500 has a mean length of 5×10^5 ym and a cross -sectional area of 2

cm". Determine the current required in the coil of 2000.turns uniformly wound ts the core to create a flux of 50/Wb in iron.

y

(1% marks)

SECTION B , ! Za

2. (a) (i) Explain what is meant by transient response. | ame 7

(ii) For the circuit in figure 1, calculate the capacitor 'pee. 5 seconds after the switch is closed. *

(2% marks)

(7 marks)

(b) Calculate the mesh currents in the circuit in figure 2, and hence find the branch currents I_1 , I_2 , and I_3 . (8 marks)

Is

(2 marks)

(8% fark

4, ci) Gi) State the Maximum Power Transfer theorem. (2 marks):

(ii) Calculate the value of R which will absorb maximum power from the circuit in figure-5. Also, compute the value of the*

maximum power, se (8 marks)

(b) Find the node voltages in the circuit in figure 6. (8 marks)

10Q, 10V, 10A

: 7 : 6Q, 10V, 10A

4 vA

; Me — 1S Vig eile

*120V 3R \$50 2 ORy

52 rr :

Figure 5 i ~ 4 10 V-

= e Ae 6

he 'SECTION Cc

; ae (a) Name three (3) types of alternating current other than sinusoidal pCa. c) waveforms and state four (4) advantages of a.c.

: es (2 marks)

(b) Use the analytical method to derive the equation for the root mean square (RMS) value of waveforms. Hence, if a sinusoidal

-waveform is defined by $V_a = 169.8 \sin(377t)$ volts, calculate the RMS voltage of the waveform, its Frequency and. 4

: instantaneous value of the voltage after a time of 6 ms. ,

7 oA

(c) State the conditions for resonance of an a.c. waveform. A circuit has a resistance of 4.0Ω and inductance of 0.50 H and

~ variable capacitance in series is connected across a 100 V , 50 Hz supply. Calculate: (i) the value of the capacitance at resonance

(ii) the voltage across the inductance and the capacitance (iii) the Q factor of the circuit. (6 marks)

= ae cam

(b) Derive the equation for the average value of waveforms. Calculate the average and effective values of the saw tooth waveform

shown in Figure 7. The waveform completes the cycle by falling back to zero instantaneously after regular interval of time.

on, é — (7 marks)

5 marks

a 8) Define the following terms, associated with the effective value of a periodic signal

© i 'Voltage V

200 V

$$i(t) = 170 \sin(628t + \phi)$$

$$v(t) = 8.5 \sin(628t + \phi)$$

Paper 3 of 5 — EEE 211 PQ with solution 2018/2019 By AKPORTU PRINCE

Level 200 • First Semester • PastQuestion

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SEMESTER: First Semester
COURSE CODE: EEE 211
TIME ALLOWED: 2 Hrs



SESSION: 2018, 2019
COURSE TITLE: Basic Electrical Engineering I
DEGREE: B.ENG

INSTRUCTION: Answer FOUR Questions from both sections

SECTION A: Answer TWO Questions from this section

1. (a) State Maximum Power Transfer theorem, and justify the need for such a theorem. (4½ marks)
- (b) (i) State Superposition theorem. Explain why this theorem does not apply to power as an electrical quantity, and state how power can be computed using this theorem. (5 marks)
- (ii) Using Superposition theorem, calculate the current flowing through the $4\ \Omega$ resistor in figure Q1 (b). Also, compute the voltage drop and power dissipated in the $4\ \Omega$ resistor. (8 marks)

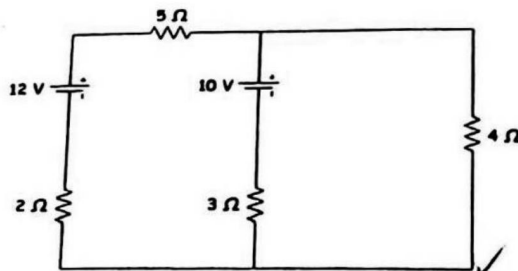


Figure Q1 (b)

2. (a) Explain the following terms: (i) Node (ii) ~~Loop~~ Branch (iv) Transient response. (6 marks)
- (b) For the circuit of figure Q2 (b), (i) Write out the differential equation (ii) Compute the time constant, natural response and forced response (3½ marks)
- (c) Calculate the mesh currents and branch currents in the circuit of figure Q2 (c). (8 marks)

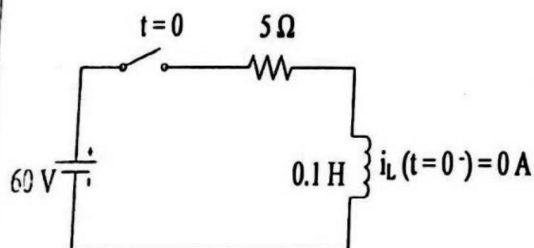


Figure Q2 (b)

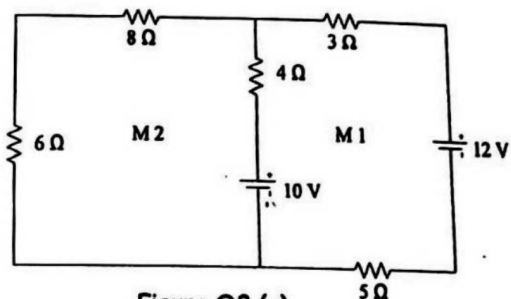


Figure Q2 (c)

3. (a) State Kirchhoff's current law and Kirchhoff's voltage law. (3 marks)
- (b) State Norton's theorem. (2½ marks)
- (c) (i) Using Thevenin theorem, calculate the current through the $8\ \Omega$ resistor in figure Q3 (c). (9 marks)
- (ii) Draw the Norton equivalent circuit of the circuit obtained in Question 3c (i) and verify the value of the current flowing through the $8\ \Omega$ resistor. (3 marks)

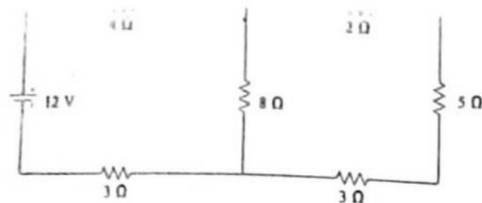


Figure Q3 (c)

SECTION B: Answer TWO Questions from this section

4. (a) State the laws of magnetism. (3½ marks)
- (b) Define the following: (i) magnetic field (ii) magnetic flux (iii) relative permeability (6 marks)
- (c) An inductive coil of resistance $10\ \Omega$ and inductance 0.1 H is connected in parallel with a $150\ \mu\text{F}$ capacitor to a variable frequency, 200 V supply. Find the resonant frequency at which the total current taken from the supply is in phase with the supply voltage. Find also the value of this current. Draw the circuit arrangement and the phasor diagrams. (8 marks)
5. (a) With the aid of a circuit diagram and all necessary equations, show that when an alternating voltage is applied to a purely capacitive circuit, the resulting current leads the applied voltage by $\frac{\pi}{2}$ radians. (8 marks)
- (b) A magnetic circuit is excited by three coils as shown in figure Q5 (b). Calculate the flux produced in the air gap. The material used for core is iron having relative permeability of 800. The length of the magnetic circuit is 100 cm with an air gap of 2 mm in it. The core has uniform cross-section of 6 cm^2 . Take $\mu_0 = 4\pi \times 10^{-7}$. (9½ marks)

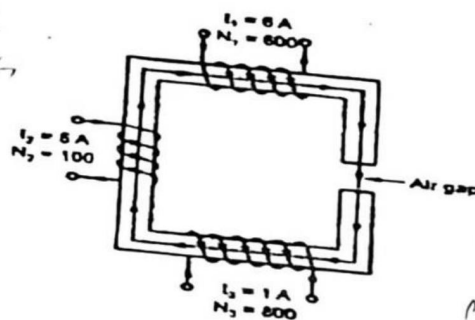


Figure Q5 (b)

6. (a) Enumerate the importance of RMS value in the field of Engineering. (6 marks)
- (b) The waveform of the voltage and current of a circuit are given by: $e = 120\sin(314t)$ and $i = 10\sin(314t + \frac{\pi}{6})$. Calculate the values of the resistance and capacitance which are connected in series to form the circuit. Also draw waveforms for current, voltage and phasor diagram. Calculate power consumed by the circuit. (11½ marks)

EEE 2018/2019 ANSWERS

a) Maximum power transfer theorem states that for any power source, the maximum power transferred from the power source to the load occurs when the resistance of the load is equal to the equivalent or input resistance.

$$R_L = R_{in}$$

NEED FOR THIS THEOREM

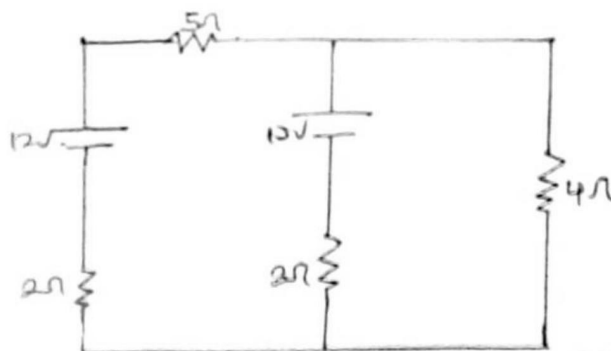
This theorem is important as it has wide applications in the analysis of communication networks because the overall efficiency of a network supplying maximum power in any branch is 50%. It is also applied in the area of improving sound quality.

Superposition Theorem

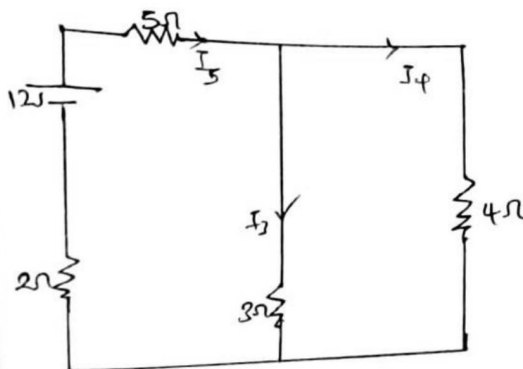
If there is more than one independent source in an electric network, the response of the network (voltage or current) can be determined by considering one source at a time. The total response is the algebraic sum of the individual responses.

Why this theorem does NOT Apply to power

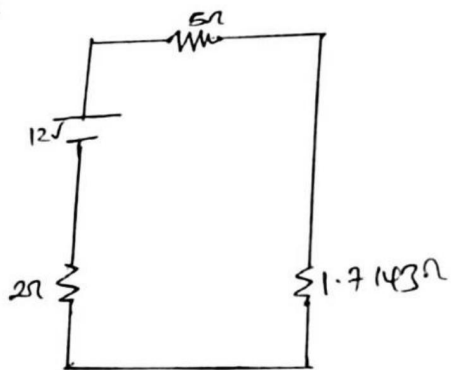
Superposition theorem is only used in linear networks, and power is not a linear network (quantity) but however it can be obtained by individual voltages and currents while making use of the formula $p = I V$ where $V = V_1 + V_2$ and $I = I_1 + I_2$.



using 12v alone.



$$R_3 // R_4 = \frac{1}{\frac{1}{3} + \frac{1}{4}}$$



$$I_5 = \frac{12}{2 + 5 + 1.7143} = \frac{12}{8.7143}$$

$$= 1.3770 \text{ A}$$

$$I_4' = \frac{3}{3+4} \times I_5$$

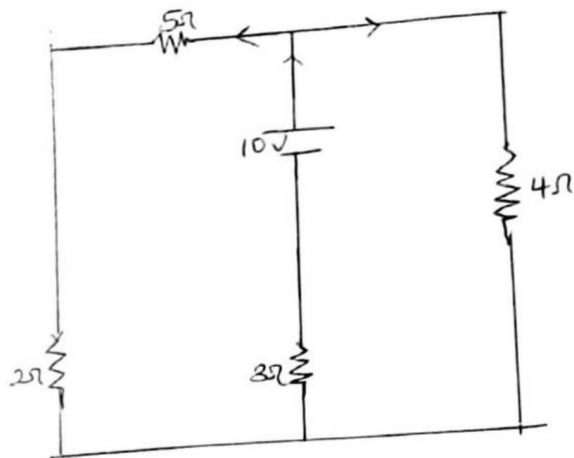
$$= \frac{3}{7} \times 1.3770$$

$$= 0.5910 \text{ A}$$

$$= 0.5910 \text{ A}$$

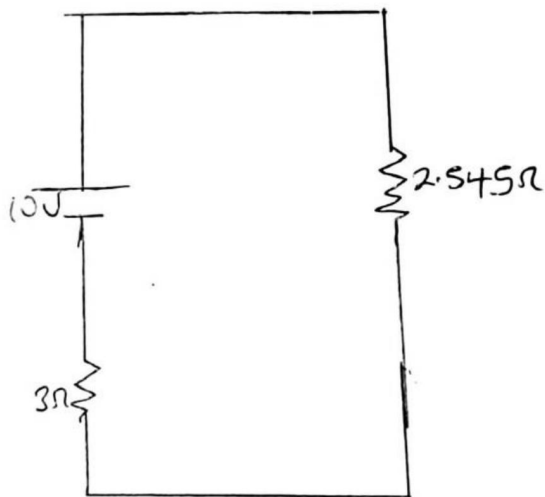
H

using 10 V alone



$$\therefore R_2 + R_5 \parallel R_4 = 7 \parallel 4$$

$$\frac{1}{7} + \frac{1}{4} = \frac{1}{R} \Rightarrow R = 2.545 \Omega$$



$$I_N = \frac{10}{3 + 2.545} = \frac{10}{5.545}$$

$$= 1.8034 \text{ A}$$

$$I_4'' = \frac{7}{7 + 4} \times 1.8034 = \frac{7}{11} \times 1.8034$$

$$= 1.1476 \text{ A}$$

S

$$I_{4\eta} = I_{4'} + I_{4''}$$

$$= 0.5901 + 1.1476$$

$$= 1.7377 \text{ A}$$

$$V_{4'} = 0.5901 \times 4 = 2.3604 \text{ V}$$

$$V_{4''} = 1.1476 \times 4 = 4.5904 \text{ V}$$

$$\text{Total } V_{4\eta} = 2.3604 + 4.5904$$

$$= 6.9508 \text{ V}$$

$$P_{4'} = 0.5901 \times 2.3604 = 1.3929 \text{ W}$$

$$P_{4''} = 1.1476 \times 4.5904 = 5.2679 \text{ W}$$

$$\text{Total } P_{4\eta} = 1.3929 + 5.2679 = 6.6608 \text{ W}$$

$$\text{Total } P_{4\eta} = 1.7377 \times 6.9508$$

$$= 12.0784 \text{ W}$$

2a)

Node:

A junction or point in a circuit where two or more circuit elements are connected together represented by a dot.

Loop

A closed path in a circuit in which no element or load is encountered more than once.

Branch

The part of a circuit that lies between two junctions.

Transient response

Transient response of a circuit to a sudden application of voltage or current

Differential equation

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{1}{L} V_s$$

$$\begin{aligned} \tau \text{ (time constant)} &= \frac{L}{R} = \frac{0.1}{5} \\ &= 0.02 \text{ s} \end{aligned}$$

$$\text{natural response} = i e^{-Rt/L} \quad \text{+ } \frac{V_s}{R}$$

$$\text{forced response} = \frac{V_s}{R}$$

$$i e^{-Rt/L} + \frac{V_s}{R} = 0 \quad (\text{Initially})$$

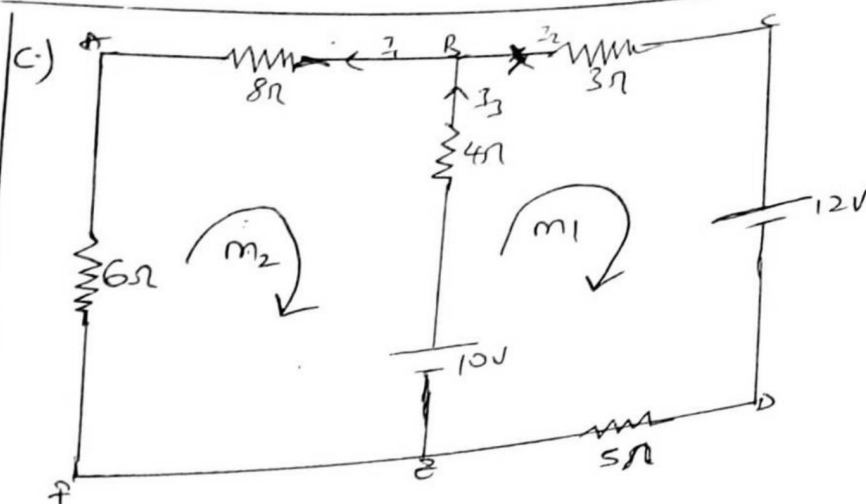
$$i e^{-\frac{5 \times 0}{0.1}} + \frac{60}{5} = 0$$

$$i e^0 + 12 = 0$$

$$i = -12$$

$$\begin{aligned} \text{natural response} &= -12 e^{-\frac{5 \times 0}{0.1}} \\ &= -12 \end{aligned}$$

$$\text{forced response} = \frac{60}{5} = 12$$



In mesh ABCEA

$$-8i_2 - 4i_2 - 10 - 6i_2 + 4i_1 = 0$$

$$-18i_2 + 4i_1 = 10 \quad \text{--- (1)}$$

In mesh BCDEB

$$-3i_1 - 12 - 5i_1 + 10 - 4i_1 + 4i_2 = 0$$

$$-12i_1 + 4i_2 = 12 \quad \text{--- (2)}$$

$$-18i_2 + 4i_1 = 10 \times 3$$

$$4i_2 - 12i_1 = 12 \times 1$$

$$2 - 5i_2 + 12i_1 = 30$$

$$+ 4i_2 - 12i_1 = 12$$

$$\hline -5i_2 = 42$$

$$-5i_2 = 42$$

$$i_2 = \frac{42}{-5}$$

$$i_2 = -0.84 \text{ A}$$

$$-18i_2 + 4i_1 = 10$$

$$-18(-0.84) + 4i_1 = 10$$

$$15.12 + 4i_1 = 10$$

$$4i_1 = -5.12$$

$$i_1 = -1.28 \text{ A}$$

$$I_1 = -(+i_2)$$

$$I_1 = i_2$$

$$I_1 = -(-0.84)$$

$$I_1 = 0.84 \text{ A}$$

$$I_2 = -1.28 \text{ A}$$

$$I_1 + I_2 = I_3$$

$$0.84 - 1.28 = I_3$$

$$I_3 = -0.44 \text{ A}$$

3A

Kirchhoff's Current Law

states that the algebraic sum of all the currents at a node is equal. i.e. the total current entering a junction or node is equal to the total current leaving.

Kirchhoff's Voltage Law

states that the algebraic sum of all voltages around any closed path or mesh equals zero.

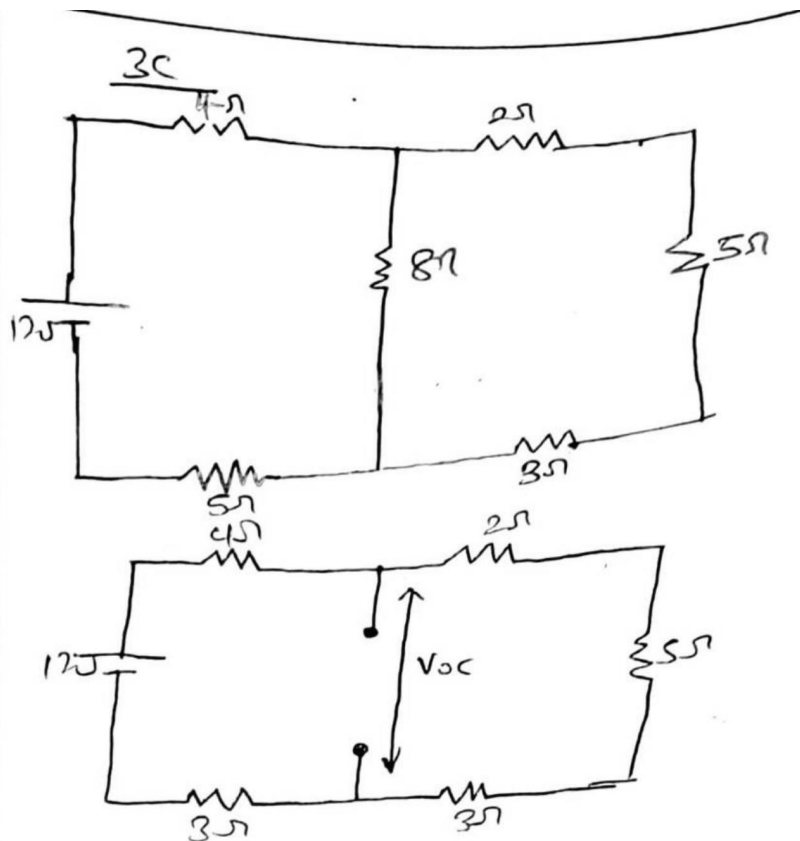
$$\sum E + \sum V_i = 0$$

3b

THEVENIN'S THEOREM

states that in any terminal active network containing voltage sources and resistance can be replaced by a constant current source in parallel with a equivalent resistor.

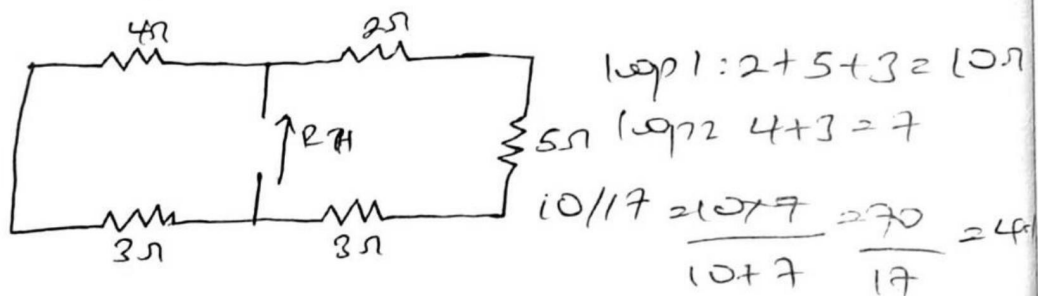
10



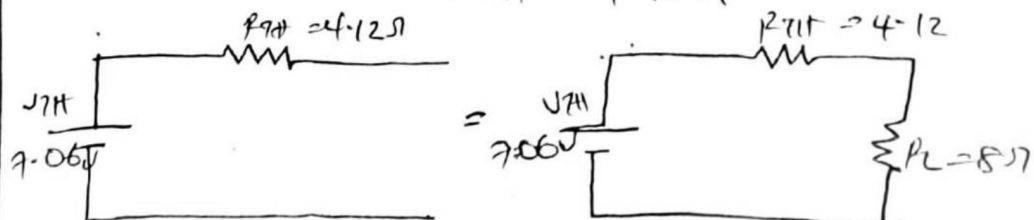
$$R = 2 + 5 + 3 = 10\Omega$$

$$\text{using } V_{DF} = \frac{10}{10+7} \times 12 = 7.06V$$

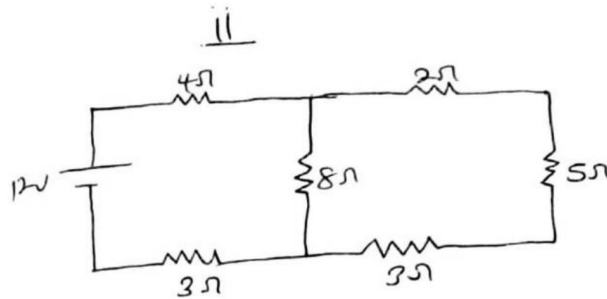
$$V_{OC} = V_{TH} = 7.06V$$



$$R_{TH} = 4.12\Omega$$



$$I_L = \frac{V_{TH}}{R_{TH} + R_L} = \frac{9.06}{4.12 + 8} = 0.585A$$



Norton equivalent circuit.

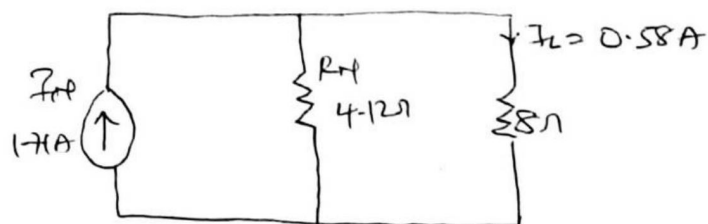
$$R_N = R_{TH} = 4.12\Omega$$

$$I_N = \frac{V_{TH}}{R_{TH}} = \frac{7.065V}{4.12\Omega} = 1.71A$$

$$I_L = \frac{R_N}{R_N + R_L} \times I_N$$

$$= \frac{4.12}{4.12 + 8} \times 1.71$$

$$= 0.58A$$



Hence it is verified using Norton's
Theorem and Thevenin's Theorem

$$I_L = 0.58 \text{ A}$$

a) i) Like poles repel one another and unlike poles attract each other

ii) The force it exerts to a magnetic material (one pole to another) is directly proportional to the product of their pole strengths and inversely proportional to the square of the distance between them. $F \propto \frac{m_1 m_2}{d^2}$ $F \propto \frac{\mu m_1 m_2}{d^2}$

4b) MAGNETIC FIELD

This is a point under the influence of a magnetic force.

MAGNETIC FLUX (ϕ)

This is the total number of lines of force existing in a particular magnetic field.

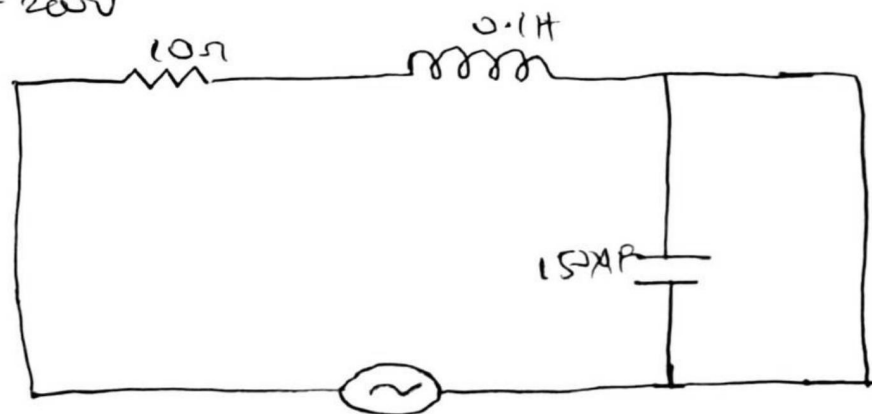
RELATIVE PERMEABILITY

This is the ratio of flux density produced in a medium other than free space to the flux density produced in free space under the influence of the same magnetic field strength and under identical conditions

$$\mu_r = B/B_0$$

4c

$R = 10\Omega$ $L = 0.1H$ $C = 150 \times 10^{-6}$
 $V = 200V$



16

Between R and L

$$Z_1 = \sqrt{R^2 + X_L^2}$$

Between R and C

$$\frac{1}{Z_2} = \sqrt{\left[\frac{1}{R}\right]^2 + \left[\frac{1}{X_C}\right]^2}$$

$$\therefore Z_T = Z_1 + Z_2$$

$$I = \frac{V}{Z_T}$$

$$X_L = 2\pi fL$$

$$2 \times 3.142 \times 41.1 \times 0.1$$

$$= 25.8 \Omega$$

$$Z_1 = \sqrt{10^2 + 25.8^2} = \sqrt{765.64} = 27.7 \Omega$$

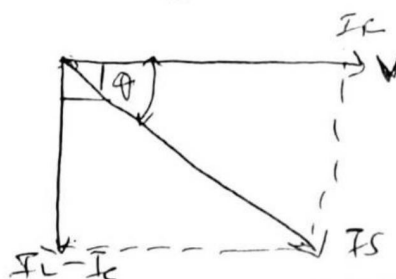
$$\frac{1}{Z_2} = \sqrt{\left(\frac{1}{10}\right)^2 + \left[\frac{1}{25.8}\right]^2} = \sqrt{0.0115023776}$$

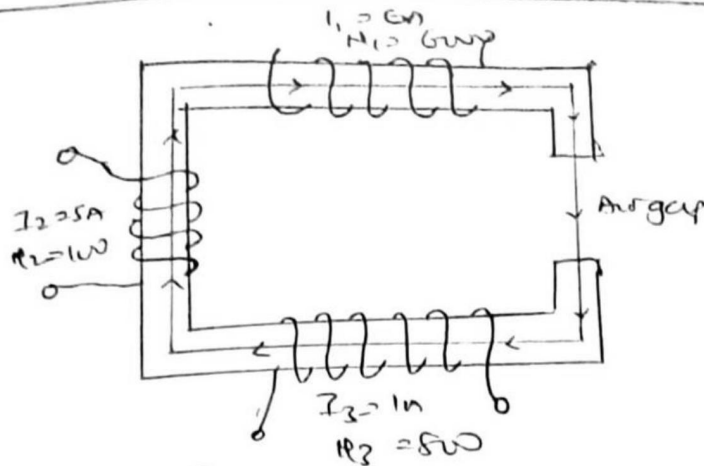
$$\frac{1}{Z_2} = 0.10725$$

$$Z_2 = \frac{1}{0.10725} = 9.32 \Omega$$

$$Z_T = 27.7 + 9.32 = 37.02$$

$$I = \frac{V}{Z_T} = \frac{200}{37.02} = 5.4 \text{ A}$$





$$N_1 = 600, I_1 = 6A, N_2 = 100, I_2 = 5A$$

$$N_3 = 800, I_3 = 1A, l = 100\text{cm} = 1\text{m}$$

$$l = l_t - l_g = 1\text{m} - 2 \times 10^{-3} = 0.998\text{m}$$

$$l_g = 2 \times 10^{-3}\text{m}, \mu_r = 2800, a = 6\text{cm}^2 = 6 \times 10^{-4}\text{m}^2$$

$$\text{Total reluctance } S = S_L + S_g$$

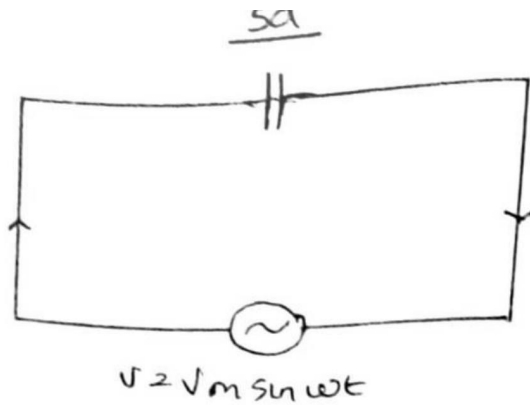
$$S_L = \frac{l}{\mu_0 \mu_r a} = \frac{0.998}{4\pi \times 10^{-7} \times 2800 \times 6 \times 10^{-4}} \\ = 1654548.263 \text{ AT/Wb}$$

To obtain flux

$$\text{Net emf} = (N_1 I_1) + (N_2 I_2) - (N_3 I_3) \\ = 600 \times 6 + 100 \times 5 - 1 \times 800 \\ NI = 3300 \text{ AT}$$

$$\phi = \frac{\text{mmf}}{\text{reluctance}} = \frac{NI}{S} = \frac{3300}{S}$$

17



$$q = CV$$

$$q = C V_m \sin \omega t$$

$$dq = I \cdot dt$$

$$I = \frac{dq}{dt} = \frac{d}{dt} (C V_m \sin \omega t)$$

$$= C V_m \omega \cos \omega t$$

$$= \frac{V_m}{\frac{1}{C \omega}} \sin(\omega t + \pi/2)$$

$$I = I_m \sin [\omega t + \pi/2]$$

$$I_m = \frac{V_m}{X_C}$$

Phase angle is positive of $+\pi/2$ i.e. $+90^\circ$, the current leads the voltage by 90° in other words the voltage lags the current by 90°

5b

16

$$S_g = \frac{I_g}{\mu_0 \mu_r} = \frac{2 \times 10^{-3}}{4\pi \times 10^{-7} \times 6 \times 10^4}$$

$$= 2652582.385 \text{ A/Wb}$$

$$S_T = S_L + S_g$$

$$= 4307130.648 \text{ A/Wb}$$

$$\phi = \frac{NI}{S_T} = \frac{3300}{4307130.648}$$

$$\phi = 0.766 \text{ mWb}$$

6A

Importance of RMS Value in the Power

ENGINEERING.

- 1) It is one of the most important parameter that is used to describe the strength of alternating current (AC).
- 2) It is equivalent to DC voltage / current that produces the same heating effect when applied across an identical resistor. It is also a measure of energy content in a given signal.
- 3) The rms value is the same as the standard deviation of the signal.
- 4) When two uncorrelated signals are added together, such as noise from two independent sources, the RMS value of their sum is equal to the square root of sum of the square of their individual RMS values.

18

6b

$$e = 120 \sin(314t)$$

$$i = 10 \sin(314t + \pi/6)$$

$$V_m = 120 \text{ V}$$

$$I_m = 10 \text{ A}$$

$$Z = \frac{V}{I} = \frac{120}{10} = 12 \text{ ohms}$$

$$\phi = \pi/6 - 0 = \pi/6 = 30^\circ$$

$$\cos \phi = R/Z$$

$$\cos 30 = 0.8660$$

$$R = Z \cos \phi$$

$$R = 12 \times 0.8660$$

$$R = 10.392 \text{ ohms}$$

$$Z^2 = R^2 + X_C^2$$

$$Z^2 = R^2 + X_C^2$$

$$X_C^2 = Z^2 - R^2$$

$$X_C = \sqrt{Z^2 - R^2}$$

$$X_C = \sqrt{12^2 - 10.392^2}$$

$$X_C = \sqrt{144 - 107.9}$$

$$X_C = \sqrt{36.1}$$

$$X_C = 6.01 \text{ ohms}$$

$$X_C = \frac{1}{2\pi fC}$$

$$I = X_C 2\pi fC$$

$$C = \frac{1}{2\pi f X_C}$$

$$C = \frac{1}{2 \times 3.142 \times f \times 6.01}$$

$$\omega = 2\pi f$$

$$f = \frac{\omega}{2\pi} = \frac{312}{3.142 \times 2}$$

$$f = \frac{312}{6.284} = 49.6 \text{ Hz}$$

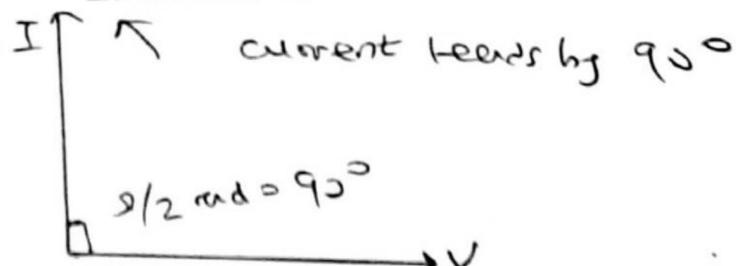
$$C = \frac{1}{2 \times 3.142 \times 49.6 \times 6.01}$$

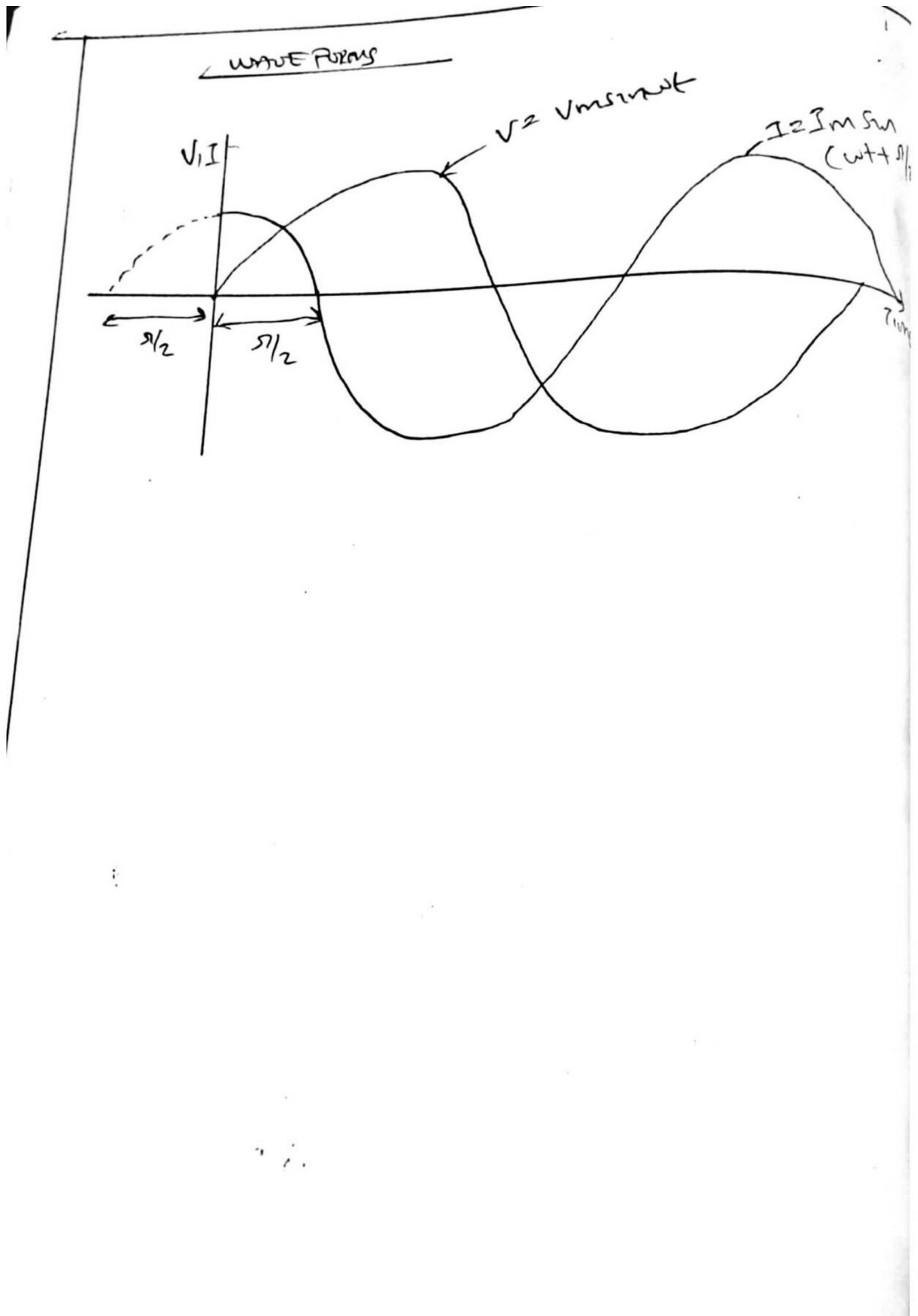
$$C = \frac{1}{1873.235264}$$

$$C = 0.0005338 \text{ F}$$

$$C = 5.34 \times 10^{-4} \text{ F}$$

PHASE DIAGRAM





Z6VLYFLC 20/24/19/2016

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY

EXAMINATION: First Semester
 COURSE CODE: EEE 211
 TIME ALLOWED: 2 ½ Hrs



SESSION: 2017/2018
 COURSE TITLE: Basic Electrical Engineering I
 DEGREE: B.ENG

INSTRUCTION: Answer FOUR Questions from both sections

SECTION A: Answer TWO Questions from this section

- (a) (i) State Superposition theorem. (3 marks)
 (ii) Using Superposition theorem, calculate the current flowing through the $8\ \Omega$ resistor in figure 1. Also, compute the voltage drop and power dissipated in the $8\ \Omega$ resistor. (7 marks)
- (b) Using Thevenin theorem, calculate the current flowing through the $12\ \Omega$ resistor in the circuit of figure 2. (7 ½ marks)

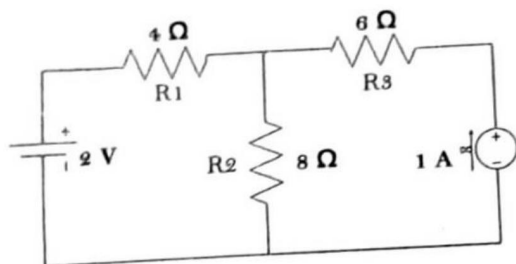


Figure 1

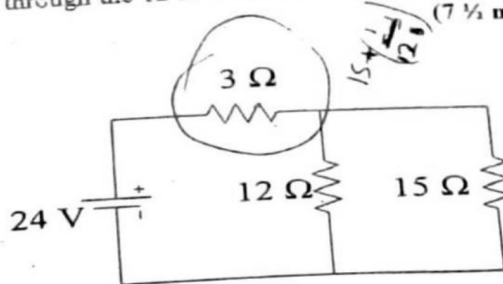


Figure 2

- (a) (i) Explain the term transient response. (3 marks)
 (ii) For the circuit of figure 3, calculate the inductor current 0.05 s after closing the switch. (7 ½ marks)
- (b) Calculate the mesh currents in the circuit of figure 4. (7 marks)

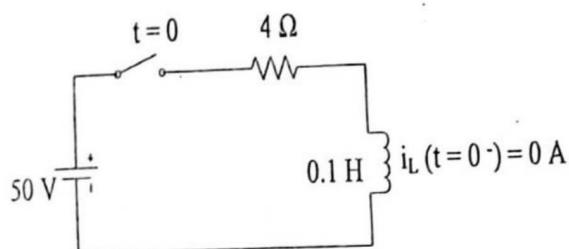


Figure 3

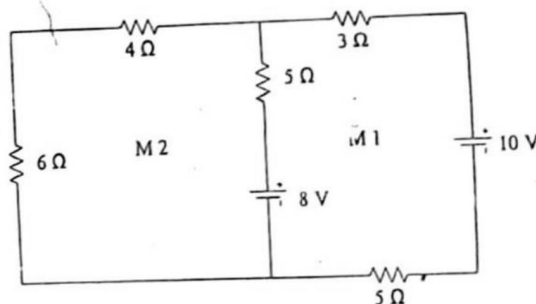


Figure 4

3. (a) (i) State Maximum Power Transfer theorem, and justify the need for such a theorem. (4 ½ marks)
 (ii) Obtain the Norton equivalent circuit and calculate the current flowing through the $8\ \Omega$ resistor in figure 5. (7 marks)
- (b) Calculate the node voltages in the circuit of figure 6 using the nodal analysis method. (6 marks)

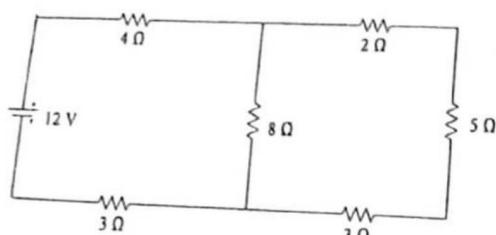


Figure 5

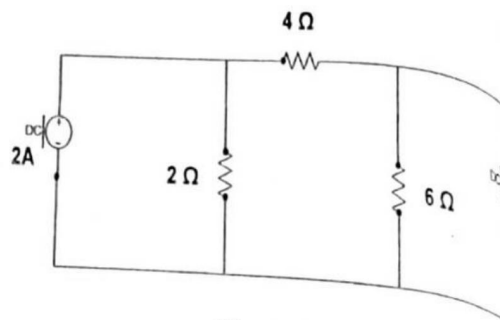


Figure 6

SECTION B: Answer TWO Questions from this section

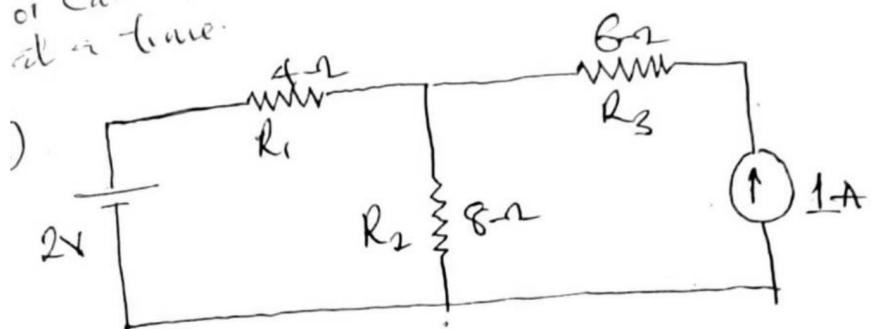
4. (a) State Coulomb's Second Law of Electrostatics. [5mks]
- (b) Define the following: (i) Electric field strength [2 ½ mks]
(ii) Electric flux [2 ½ mks]
- (c) A charge of $1.5 \mu\text{C}$ is carried on two parallel rectangular plates each measuring 60 mm by 80 mm. Calculate the electric flux density. If the plates are spaced 10mm apart and voltage between them is 0.2 kV, determine the electric field strength. [Take $\mu_0 = 4\pi \times 10^{-7}$] [7mks]
5. (a) Define the following: (i) Reluctance [2 ½ mks] (ii) m.m.f. [2 ½ mks]
- (b) An iron ring 8cm mean diameter is made up of round iron of diameter 1cm and permeability of 900. It consists of winding with 400 turns carrying a current of 3.5 A. Determine:
(i) m.m.f. [3 ½ mks] (ii) total reluctance [3 ½ mks] (iii) the flux [1 ½ mk]
(iv) flux density in the ring [1 ½ mk] (v) Draw the magnetic circuit [2 ½ mks]
6. (a) With the aid of a circuit diagram and all necessary equations show that for a purely inductive circuit the instantaneous current is given by $i = I_m \sin(\omega t - \frac{\pi}{2})$. [8 ½ mks]
- (b) A voltage $e = 200 \sin 100\pi t$ is applied to a load having $R = 200\Omega$ in series with $L = 638\text{mH}$. Find:
(i) An expression for the instantaneous current [4 ½ mks]
(ii) the power consumed by the load [1 ½ mks]
(iii) the reactive power of the load [1 mk]
(iv) the voltage across R and L [1 mk]
(v) Draw the circuit diagram [1 ½ mks]

EEE 211 { BASIC ELECTRICAL ENGINEERING }

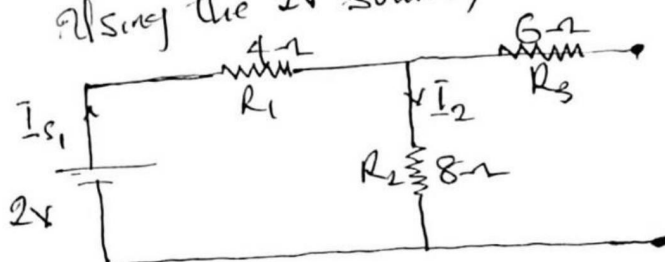
2017/2018

Question 1

1) If there is more than one independent source in an electric network, the response of the network (i.e. Voltage, or Current) can be determined by considering one source at a time.

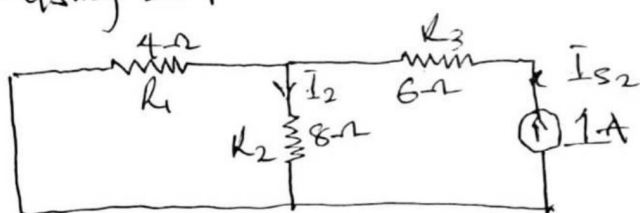


Using the 2V Source,



$$I_{S1} = I_2 = \frac{2}{8+4} = \frac{2}{12} = \frac{1}{6} = 0.167A$$

Using 1A Source



Using Current Divider Rule.

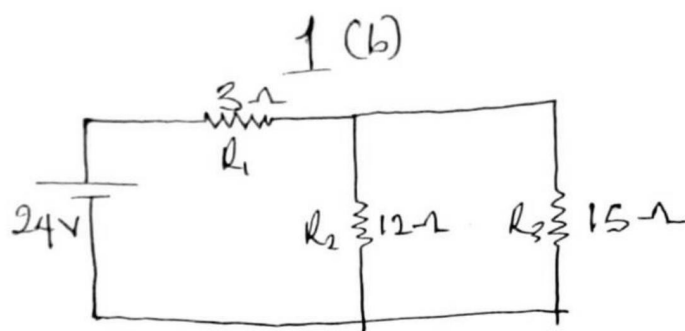
$$I_2 = \frac{R_1}{R_1 + R_2} \times I_{S2} = \frac{4}{4+8} \times 1 = \frac{4}{12} = 0.33A$$

2H

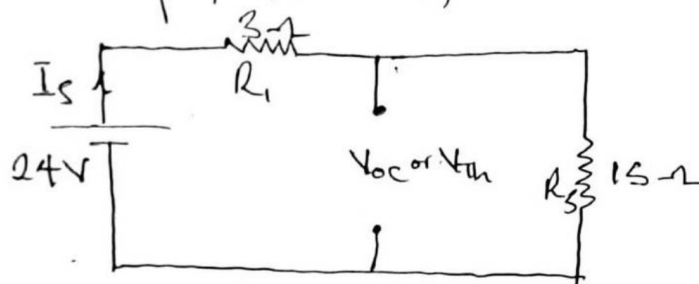
Using both sources, $I_T = 0.167 + 0.33$
 $= 0.497 \approx 0.50 \text{ A}$

Voltage Drop, $V_{R_2} = I_T \times R_2$
 $= 0.50 \times 8$
 $= 4.0 \text{ V}$

Power, $P = I_T V_{R_2} = 0.50 \times 4$
 $= 2 \text{ W}$



To find V_{oc} or V_{th} , we remove the 12Ω Resistor



$$I_S = \frac{V}{R_1 + R_3} = \frac{24}{15 + 3} = \frac{24}{18} = 1.33 \text{ A}$$

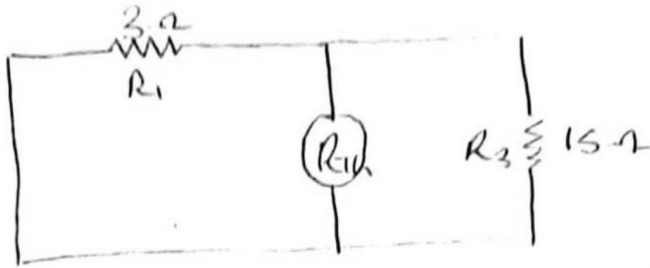
$$V_{oc} \text{ or } V_{th} = I_S \times R_3$$

$$= 1.33 \times 15$$

$$= 19.95 \text{ V}$$

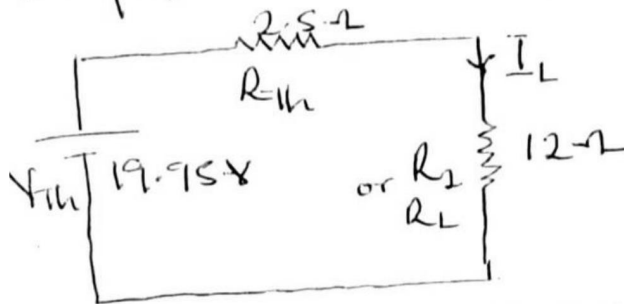
Also, To find R_{th} , we remove the Source Not

25



$$R_{th} = R_1 // R_3 = \frac{R_1 \times R_3}{R_1 + R_3} = \frac{3 \times 15}{3 + 15} = \frac{45}{18} = 2.5 \Omega$$

Therefore, the Thevenin Equivalent Circuit will be



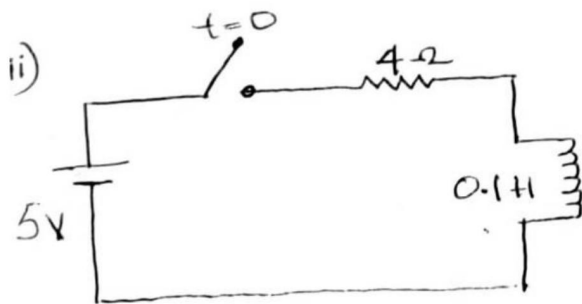
$$\therefore I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{19.95}{2.5 + 12} = \frac{19.95}{14.5} = 1.38 A$$

26.

Question 2

i)

Transient response is the response of a circuit to the sudden application of voltage or current. Transient response occurs in circuits that energy storage element (e.g. Inductors & Capacitors)



$$\text{At } t=0, i_L = 0 \text{ A}$$

$$\frac{di_L(t)}{dt} + \frac{R i_L(t)}{L} = \frac{V_s}{L}$$

$$\frac{L}{R} = \frac{0.1}{4} = 0.025 \text{ s}$$

$$i_{LN}(t) = K e^{-t/\tau} = K e^{-t/0.025}$$

$$i_{Lf}(t) = \frac{V_s}{R} = \frac{50}{4} = 12.5$$

$$\therefore i_L(t) = i_{LN}(t) + i_{Lf}(t)$$

$$\text{At } t=0$$

$$i_L(0) = K e^{-0/0.025} + 12.5$$

$$0 = K e^0 + 12.5$$

27

$$0 = k + 12.5 \quad ; \quad k = -12.5$$

$$\therefore i_L(t) = -12.5 e^{-t(0.025 + 12.5)}$$

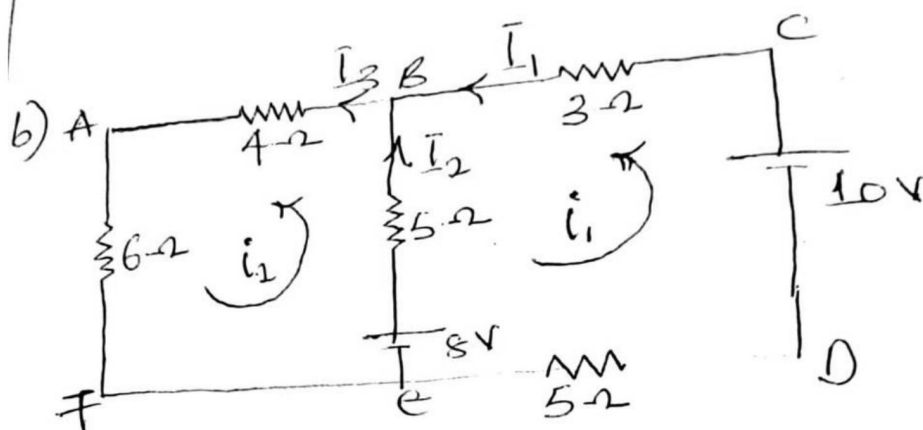
$$\text{At } t = 0.05 \text{ secs}$$

$$i_L(0.05) = -12.5 e^{-0.05(0.025 + 12.5)}$$

$$= -12.5 e^{-2} + 12.5$$

$$= -1.69 + 12.5$$

$$i_L(0.05) = \underline{\underline{10.81 \text{ A}}}$$



Note: $I_1 = i_1$
 $I_2 = i_2 = I_1 + I_2$
 $I_2 = I_3 - I_1 = i_2 - i_1$

Taking Mesh BEDCB

$$10 - 3I_1 + 5I_2 - 8 - 5I_1 = 0$$

$$10 - 8 - 3i_1 + 5(i_2 - i_1) - 5i_1 = 0$$

$$2 - 3i_1 + 5i_2 - 5i_1 - 5i_1 = 0$$

$$+5i_2 - 13i_1 + 2 = 0$$

$$13i_1 - 5i_2 = 2 \quad \text{--- --- --- --- --- (1)}$$

Also, Taking Mesh BAFEB

$$8 - 5I_2 - 4I_3 - 6I_3 = 0$$

$$8 - 5(i_2 - i_1) - 4i_2 - 6i_2 = 0$$

$$8 - 5i_2 + 5i_1 - 4i_2 - 6i_2 = 0$$

$$8 - 15i_2 + 5i_1 = 0$$

$$5i_1 - 15i_2 = -8 \quad \text{--- (2)}$$

Solving equations (1) and (2) simultaneously

$$13i_1 - 5i_2 = 2 \quad \text{--- (1) } \times 5$$

$$5i_1 - 15i_2 = -8 \quad \text{--- (2) } \times 13$$

$$65i_1 - 25i_2 = 10$$

$$65i_1 - 195i_2 = -104$$

$$170i_2 = 114$$

$$i_2 = \frac{114}{170} = 0.67 \text{ A}$$

Substituting $i_2 = 0.67$ into eqn (1) to find i_1

$$13i_1 - 5i_2 = 2 \quad ; \quad 13i_1 - 5(0.67) = 2$$

$$i_1 = \frac{5.35}{13} = 0.41 \text{ A}$$

$$\therefore i_1 = 0.41 \text{ A}$$

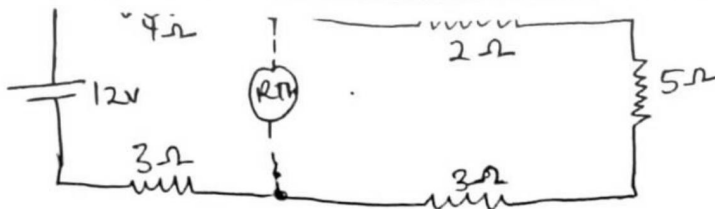
$$i_2 = 0.67 \text{ A}$$

$$I_1 = 0.41 \text{ A}$$

$$I_3 = 0.67 \text{ A}$$

$$I_2 = I_3 - I_1 = 0.67 - 0.41 = 0.26 \text{ A}$$

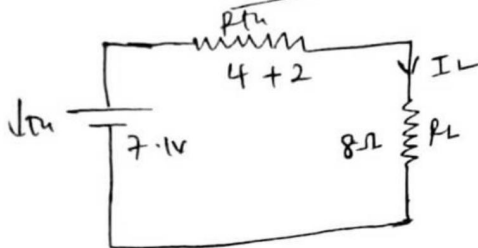
29



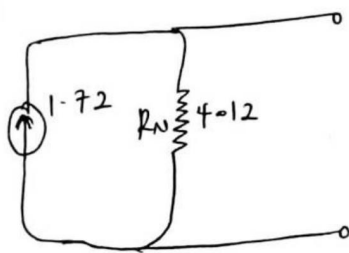
$$R_{th} = 4 + 3 \parallel 2 + 3 + 5$$

$$= 7 \parallel 10$$

$$= 4.12 \Omega$$

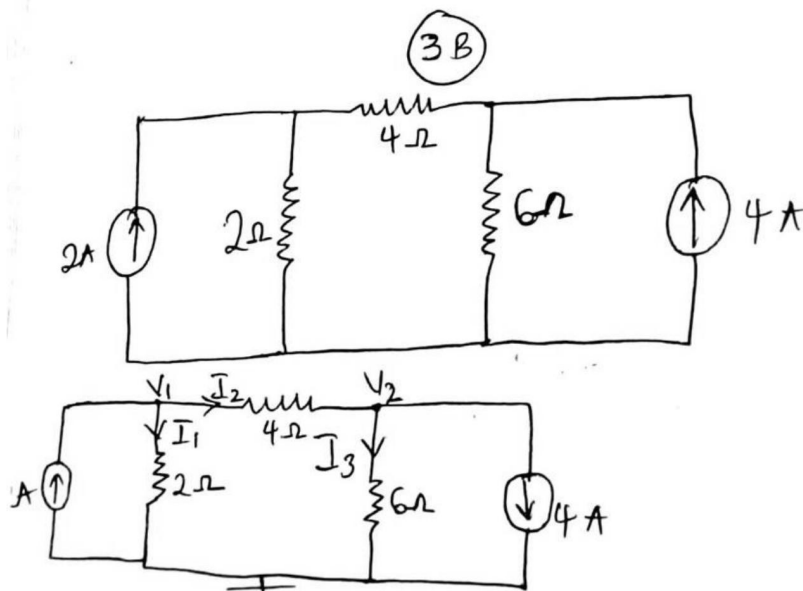


$$I_L = \frac{7.1}{4.12 + 8} = \frac{7.1}{12.12} = 0.59 A$$



$$V_{th} = I_N R_N$$

$$I_N = \frac{V_{th}}{R_N} = \frac{7.1}{4.12} = 1.72$$



Using KCL at node 1

$$2 = \bar{I}_1 + \bar{I}_2$$

$$\bar{I}_1 = \frac{V_1}{2}, \quad \bar{I}_2 = \frac{V_1 - V_2}{4}$$

$$2 = \frac{V_1}{2} + \frac{V_1 - V_2}{4}$$

$$2 = \frac{2V_1 + V_1 - V_2}{4}$$

$$8 = 2V_1 + V_1 - V_2$$

$$8 = 3V_1 - V_2 \quad \text{--- (1)}$$

Using KCL at node 2

$$\bar{I}_2 = \bar{I}_3 + 4$$

$$\bar{I}_3 = \frac{V_2}{6} \quad \bar{I}_2 = \frac{V_1 - V_2}{4}$$

$$\frac{V_1 - V_2}{4} = \frac{V_2}{6} + 4$$

$$\frac{V_1 - V_2}{4} = \frac{V_2 + 24}{6}$$

$$6V_1 - 6V_2 = 4V_2 + 96$$

$$6V_1 - 6V_2 - 4V_2 = 96$$

$$6V_1 - 10V_2 = 96 \quad \text{--- (2)}$$

$$2 \times 3V_1 - V_2 = 8$$

$$1 \times 6V_1 - 10V_2 = 96$$

$$1 \times 6V_1 - 10V_2 = 96$$

$$6V_1 - 2V_2 = 16$$

$$-6V_1 - 10V_2 = 96$$

$$\hline 8V_2 = -80$$

3

$$\begin{aligned}
 V_2 &= -10\text{V} \\
 6V_1 - 10V_2 &= 96 \\
 6V_1 - 10(-10) &= 96 \\
 6V_1 + 100 &= 96 \\
 6V_1 &= 96 - 100 \\
 6V_1 &= -196 \\
 V_1 &= \frac{-196}{6} \\
 V_1 &= -32.67\text{V}
 \end{aligned}$$

(4A)

Coulomb's second law of Electrostatic states that the force of attraction between two charges is inversely proportional to the square of the distance and directly proportional to the charges.

(4b)i

Electric field strength is the force which would be experienced by a particle having charge Q when placed in the field $E = F/Q$

(4b)ii

Electric flux is the total electric field coming from a charged body.

(5a)i

Reluctance is the resistance offered by the material to the flow of magnetic flux through it. It is

$$S_i = \frac{0.249}{4 \times \pi \times 10^{-7} \times 950 \times 7.854 \times 10^{-3}}$$

$$S_i = \frac{0.249}{8.883 \times 10^{-8}}$$

$$S_i = 2803219.414 \text{ A}\cdot\text{T}/\text{wb} = 2.80321944 \times 10^6 \text{ A}\cdot\text{T}/\text{wb}.$$

$$\begin{aligned} S_g &= \frac{L_g}{\mu_0 \mu_r g} = \frac{0.002}{4\pi \times 10^{-7} \times 1 \times 7.854 \times 10^{-5}} \\ &= 20264189.34 \\ &= 20.26418934 \times 10^6 \text{ A}\cdot\text{T}/\text{wb}. \end{aligned}$$

$$S_T = 2.803219414 \times 10^6 + 20.26418934 \times 10^6$$

$$S_T = 23.067 \times 10^6 \text{ A}\cdot\text{T}/\text{wb}.$$

$$\begin{aligned} \text{Flux } \phi &= \frac{\text{mmf}}{\text{reluctance}} = \frac{NI}{S_T} = \frac{1400}{23.067 \times 10^6} \\ &= 6.069 \times 10^{-5} \text{ wb} \end{aligned}$$

$$\begin{aligned} \text{Flux density} &= \frac{\phi}{a} = \frac{6.069 \times 10^{-5}}{7.854 \times 10^{-5}} \\ &= 0.7727 \text{ wb/m}^2 \end{aligned}$$

34

denoted by 'S'.

$$\text{Reluctance } S = \frac{NI}{I}$$

(5a) ii

m.m.f (magnetomotive force) is the work done in joules on a unit magnetic pole in taking it round a closed magnetic circuit. ∴ m.m.f

(5b)

$$\text{Diameter of ring} = 8\text{cm} = 0.08\text{m} = 8 \times 10^{-2}$$

$$\mu_r = 900$$

$$L_g = 2\text{mm} = 0.002\text{m} = 2 \times 10^{-3}\text{m}$$

$$\mu_0 = 4\pi \times 10^{-7}$$

$$N = 400$$

$$I = 3.5\text{A}$$

$$\text{Diameter of iron} = 1\text{cm} = 0.01\text{m}$$

$$\text{Area} = \frac{\pi d^2}{4} = \frac{\pi \times (0.01)^2}{4} = 7.85 \times 10^{-5}$$

$$\text{(i) } \text{mmf} = NI = 400 \times 3.5 = 1400\text{ AT}$$

$$\text{(ii) Total reluctance } S_T = S_i + S_g$$

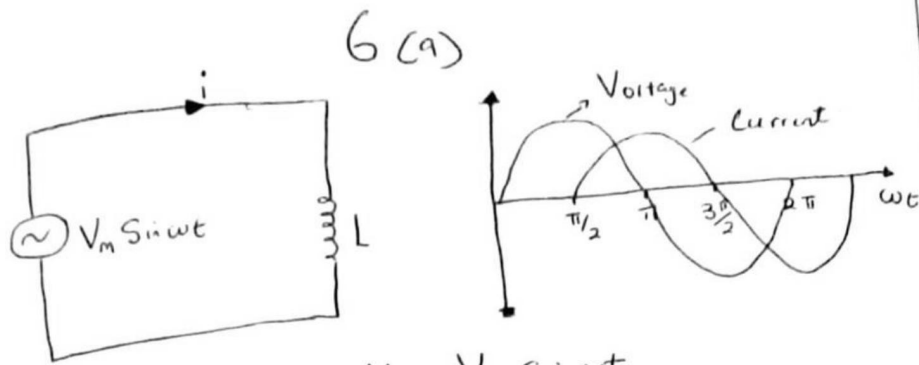
$$S_i = \frac{L_i}{\mu_0 \mu_r a}$$

$$\text{Length of iron} = \pi d - \text{length of air gap}$$

$$= \pi \times 8 \times 10^{-2} - 2 \times 10^{-3}$$

$$L_i = 0.249\text{m}$$

35



Let input Voltage $V = V_m \sin \omega t$
 Recau; The Voltage drop in an inductor $= V_L = L \frac{di}{dt}$

$$\therefore L \frac{di}{dt} = V_m \sin \omega t$$

$$di = \frac{V_m}{L} \sin \omega t dt$$

Integrating both Side of the eqn. w.r.t

$$\int di = \int \frac{V_m}{L} \sin \omega t dt$$

$$i = \frac{V_m}{L} \int \sin \omega t dt$$

$$i = \frac{V_m}{\omega L} (-\cos \omega t)$$

Recau; $\cos x = \sin(90 - x) \therefore \text{MSO; } \pi/2 \text{ ms} = 90^\circ$
 $\cos(-x) = \sin(90 - x)$

So; $i = \frac{V_m}{\omega L} \sin(90 - \omega t)$

$$i = \frac{V_m}{\omega L} \sin(\omega t - \pi/2)$$

3b

6(b)

Given; $e = 200 \sin 100\pi t$
 $R = 200 \Omega$ and $L = 638 \text{ mH}$

(i)

By comparing $200 \sin 100\pi t$ to $E_m \sin \omega t$

$$E_m = 200 \quad \omega = 100\pi \quad f = 50 \text{ Hz}$$

Also; $Z = R + jX_L = 200 + j200.46$

Where. $X_L = 2\pi fL$ or $\omega L = 100\pi (638 \text{ mH})$

$$X_L = 200.46 \Omega$$

$$|Z| = \sqrt{R^2 + (X_L)^2} = \sqrt{200^2 + (200.46)^2}$$

$$|Z| = 283.17 \Omega$$

Recall;

$$i = \frac{e \text{ or } V}{Z} = \frac{200 \sin 100\pi t}{283.17}$$

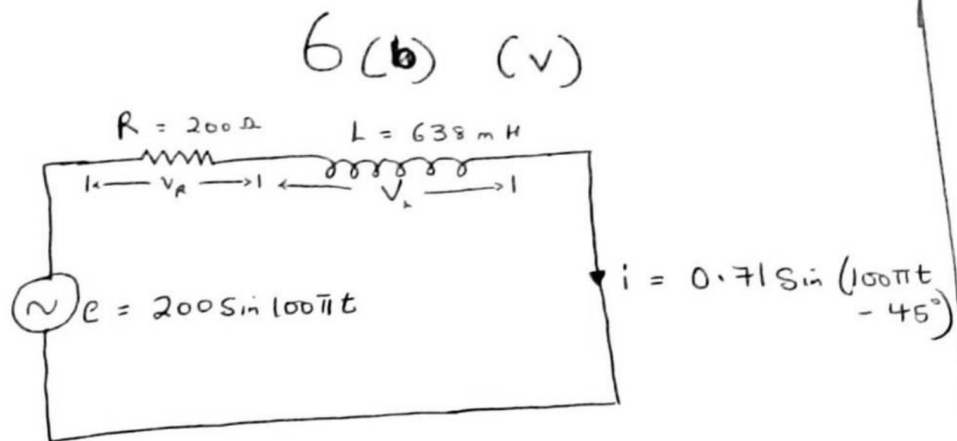
$$i = 0.71 \sin(100\pi t - \phi)$$

Since $\underline{I}$ lags behind Voltage

Where $\phi = \tan^{-1}\left(\frac{X_L}{R}\right) = \tan^{-1}\left(\frac{200.46}{200}\right)$
 $\phi = 45^\circ$

$$\therefore i = 0.71 \sin(100\pi t - 45^\circ)$$

or $\frac{\pi}{4}$



6b(ii)

POWER CONSUMED BY LOAD (R)

$$P_R = I_{\text{r.m.s}}^2 \times R$$

$$= \left(\frac{0.71}{\sqrt{2}} \right)^2 \times 200$$

$$P_R = 50.41 \text{ W}$$

6b(iii)

REACTIVE POWER

$$Q_L = I^2 X_L \text{ or } IV \sin \phi$$

$$= \frac{0.71}{\sqrt{2}} \times \frac{200}{\sqrt{2}} \times \sin 45^\circ$$

$$Q_L = 50.20 \text{ W}$$

38

(iv)

Voltage drop Across R

$$V_R = I_{r.m.s} \times R$$

$$= \frac{0.71}{\sqrt{2}} \angle -45^\circ \times 200 \angle 0^\circ$$

$$= \underline{\underline{100 \angle -45^\circ \text{ V}}}$$

Voltage drop Across L

$$V_L = I_{r.m.s} \times X_L$$

$$= \frac{0.71}{\sqrt{2}} \angle -45^\circ \times 200.46^\circ \angle 90^\circ$$

$$\angle 90^\circ = jX_L$$

$$= \underline{\underline{100.23 \angle 45^\circ \text{ V}}}$$

Phase Angle Remains 90°
 -45° to 45°

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY

EXAMINATION: First Semester
COURSE CODE: EEE 211



SESSION: 2016/2017
COURSE TITLE: Basic Electrical
Engineering I
DEGREE: B.ENG

TIME ALLOWED: 2Hrs 30mins

INSTRUCTION: Answer FOUR Questions from both sections

SECTION A: Answer TWO Questions from this section

1. (a) State Kirchhoff's current law and Kirchhoff's voltage law. (3 marks)
- (b) (i) Obtain the node voltages in the circuit of figure 1 using nodal analysis. (7 marks)
- (ii) Using Superposition theorem, calculate the current flowing through the $4\ \Omega$ resistor in figure 2. (7 ½ marks)

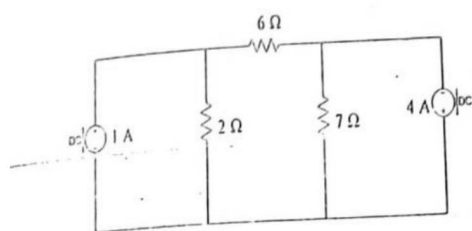


Figure 1

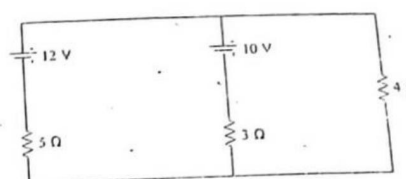


Figure 2

2. (a) Explain the term transient response. (2 marks)
- (b) (i) Calculate the mesh currents in the circuit of figure 3. (7 marks)
- (ii) In the circuit of figure 4, find the value of R_L necessary for maximum power transfer, and hence calculate the maximum power that can be delivered to R_L . (8 ½ marks)

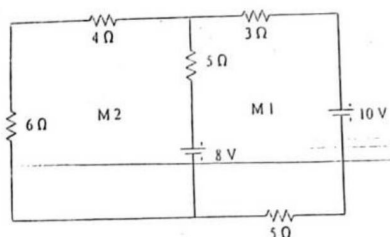


Figure 3

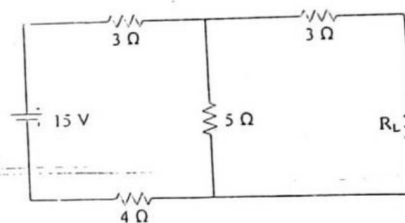


Figure 4

40

3. (a) State Thevenin theorem and Norton theorem.

(5 marks)

(b) (i) Obtain the Thevenin equivalent circuit and calculate the current flowing through the $8\ \Omega$ resistor in figure 5.

(9 marks)

(ii) Draw the Norton equivalent circuit of the circuit obtained in Question 3b(i) and verify the value of current flowing through the $8\ \Omega$ resistor.

(3 1/2 marks)

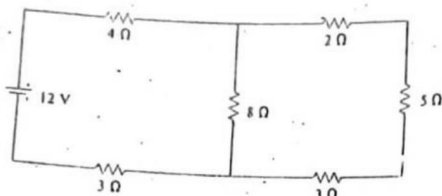


Figure 5

SECTION B: Answer TWO Questions from this section

4. (a) Define briefly the following (i) Magneto motive force (mmf) (ii) Magnetic flux density.

(b) Three concentric spheres of radii 4, 6, and 8 cm have charges of +8, -6 and +4 μC respectively. What are the potentials and field strengths at points 2 cm, 5 cm, 7 cm and 10 cm from the centre? (17 1/2 marks)

5. (a) List the methods used in analyzing alternating current (ac).

(b) An alternating voltage is given by $v = 75 \sin(200\pi t - 0.25)$ volts. Find (i) the amplitude, (ii) the peak value, (iii) the r.m.s. value, (iv) the periodic time, (v) the frequency, and (vi) the phase angle (in degrees and minutes) relative to $75 \sin 200\pi t$. (17 1/2 marks)

6. (a) Define the following: (i) Apparent power (ii) Power factor.

(b) (i) A load takes 90 kW at a power factor of 0.5 lagging. Calculate the apparent power and the reactive power.

(ii) A series circuit of resistance $60\ \Omega$ and inductance 75 mH is connected to a 110 V, 60 Hz supply. Calculate the power dissipated. (17 1/2 marks)

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41

2016 / 2017 SESSION

ANSWERS

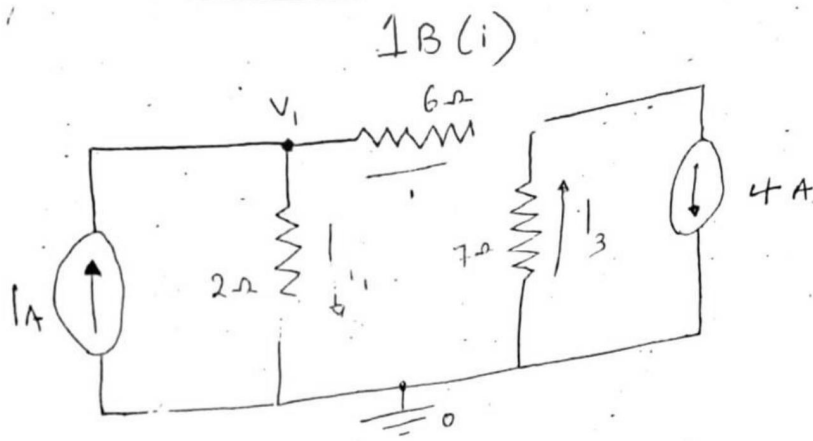
1(A)

2014 / 2015

SESSION

Q. 3 (A)

~~8~~ 42



Applying KCL at Node 1

$$1 = I_1 + I_2 \quad \text{--- (1)}$$

$$1 = \frac{V_1 - 0}{2} + \frac{V_1 - V_2}{6}$$

$$6 = 3V_1 + V_1 - V_2$$

$$4V_1 - V_2 = 6 \quad \text{--- (2)}$$

Applying KCL at Node 2

$$4 = I_2 + I_3 \quad \text{--- (3)}$$

$$4 = \frac{V_1 - V_2}{6} + \frac{0 - V_2}{7}$$

$$168 = 7V_1 - 7V_2 - 6V_2$$

$$7V_1 - 13V_2 = 168 \quad \text{--- (4)}$$

~~84~~

1 B (i)

Solving Equation (2) & (4) Simultaneously

$$7V_1 - 13V_2 = 168 \quad \text{--- (4)}$$

$$- \boxed{4V_1 - V_2 = 6} \quad \text{--- (2) } \times 13$$

$$52V_1 - 13V_2 = 78 \quad \text{--- (5)}$$

$$\underline{\hspace{1cm}} - 45V_1 = 90$$

$$V_1 = -\frac{90}{45} = -2V$$

Substituting the value of V_1 in eqn (2)

$$4(-2) - V_2 = 6$$

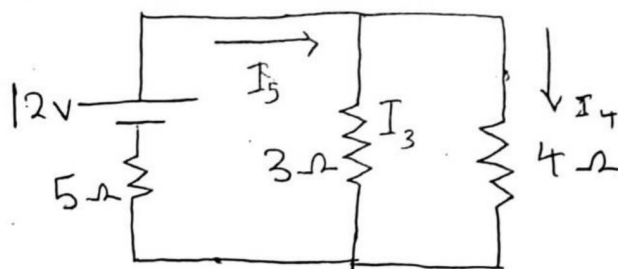
$$-8 - V_2 = 6$$

$$V_2 = -14V$$

1 B (ii)

Applying Superposition theorem

Considering 12v Source only

~~8.5~~

44

$$I_5 = \frac{V}{[5 + (3//4)] \Omega}$$

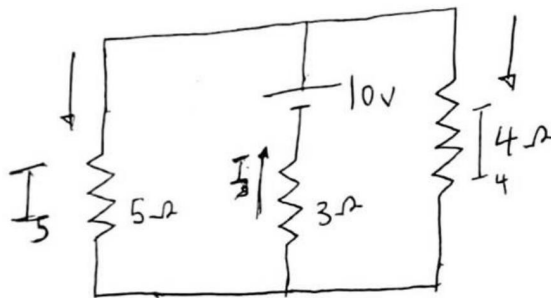
$$= \frac{12}{5 + 6.714} = \frac{12}{6.714}$$

$$= 1.787$$

$$I_4 = \frac{3}{3+4} \times 1.787 \text{ (KCL)}$$

$$I_4' = 0.766 \text{ Amps}$$

CONSIDERING 10V Source Only



$$I_3 = \frac{10V}{[3 + (5//4)] \Omega} = 1.915 \text{ Amps}$$

$$I_4 = \frac{5}{9} \times 1.915 \text{ (KCL)}$$

~~8.6~~

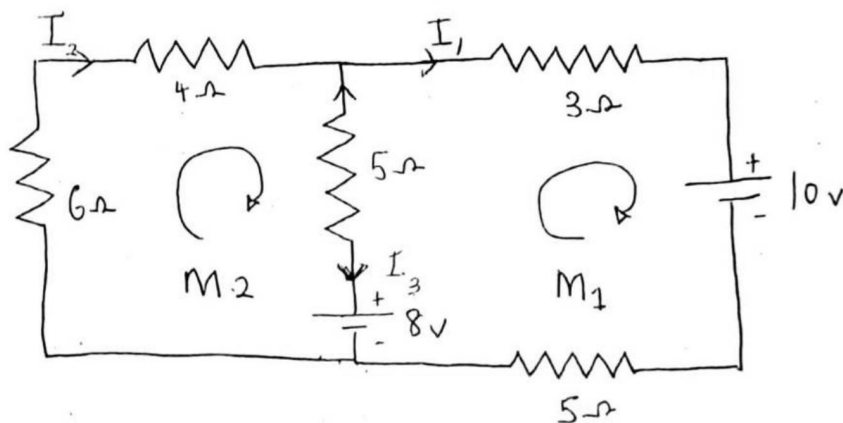
2(A)

2015/2016

SESSION

Q. 2(A)

2B (i)



Apply KVL in both MESH
moving Clockwise in both MESH



~~7~~ $I_4'' = 1.064$

$$I_4 = I_4' + I_4''$$
$$= 0.766 + 1.064$$
$$= \underline{1.832} \text{ Amps } \underline{1.84A}$$

MESH 2

$$-6i_2 - 4i_2 - 5i_2 + 5i_1 - 8 = 0$$

$$-15i_2 + 5i_1 = 8$$

$$5i_1 - 15i_2 = 8 \quad \text{--- (1)}$$

MESH 1

$$8 - 10 - 5i_1 - 3i_1 - 5i_1 + 5i_2 = 0$$

$$-2 - 13i_1 + 5i_2 = 0$$

$$13i_1 - 5i_2 = -2 \quad \text{--- (2)}$$

Solving Eqn (1) & Eqn (2)

Simultaneously.

$$5i_1 - 15i_2 = 8 \quad \text{--- (1)}$$

$$39i_1 - 15i_2 = -6 \quad \text{--- (3)}$$

$$\text{Eqn (3)} = \text{Eqn (2)} \times 3$$

$$i_1 = -0.412 \text{ Amps}$$

$$i_2 = -0.671 \text{ Amps}$$

$$-34i_1 = 14$$

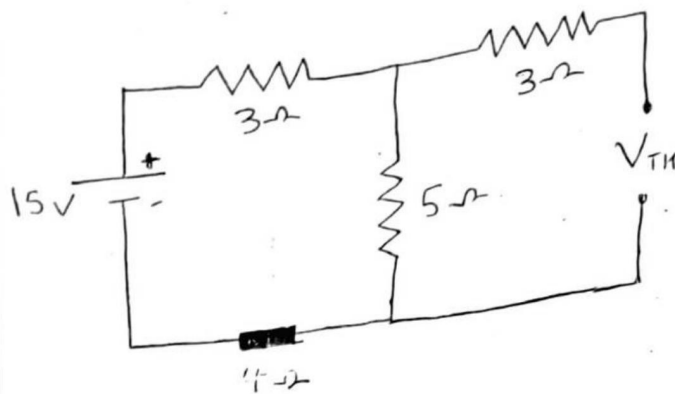
$$i_1 = \frac{14}{-34}$$

48

2B(i)

N.B -ve Sign implies the reverse
is the case of current flow.

2B(ii)



Applying KVL

$$V_{TH} = \frac{5}{4+3+5} \times 15V = \frac{5}{12} \times 15$$

$$= 6.25V$$

$$R_L = R_{TH} \text{ at Maximum Power}$$

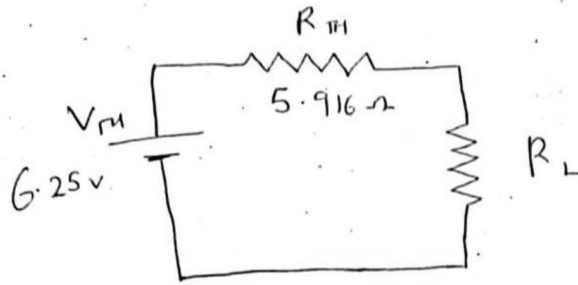
$$= [(4+3) \parallel 5 + 3] \Omega$$

49

49

2 B (ii)

$$R_L = R_{TH} = 5.916 \, \Omega$$



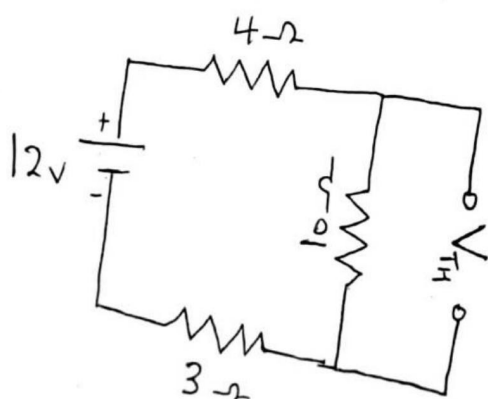
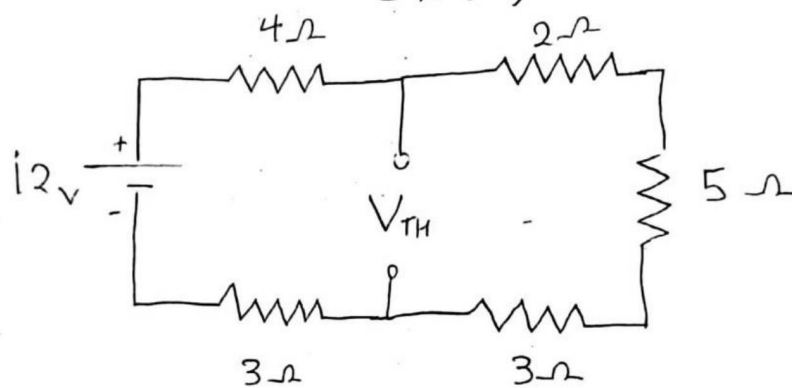
$$P_{max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{6.25^2}{4 \times 5.916}$$

$$P_{max} = 1.651 \text{ watt}$$

SD

~~SD~~

3(A)
 Thevenin theorem -7. 2013/2014 Q. 2(n)
NORTON THEOREM: It states that
 any linear circuit containing several
 energy sources and resistances can
 be replaced by a single constant
 current generator in parallel with
 a single resistor.
 3B(i)



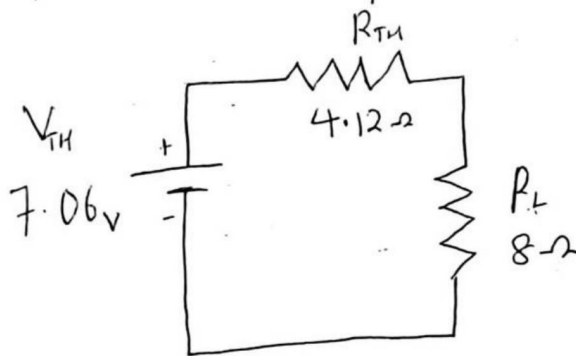
Q

$$V_{TH} = \frac{10}{4+3+17} \times 12$$

$$V_{TH} = 7.06 \text{ Volts.}$$

$$R_{TH} = \frac{10 \times 7}{17} = 4.12 \text{ ohms}$$

Thevenin's Equivalent Circuit



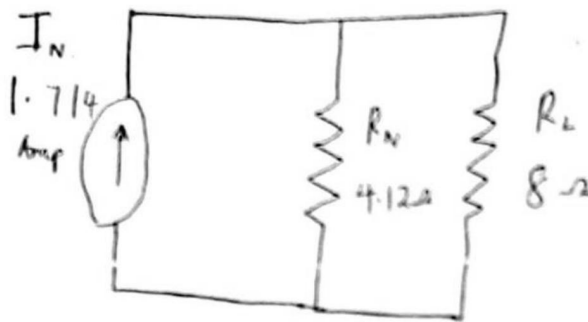
$$I_{8\Omega} = \frac{V_{TH}}{R_{TH} + R_L} = \frac{7.06}{4.12 + 8} = 0.583 \text{ Amps}$$

~~52~~ 52.

3 B (11)
Norton Equivalent Circuit

$$I_N = \frac{V_{TH}}{R_{TH}} = \frac{7.06}{4.12} = 1.714 \text{ A}$$

$$R_{TH} = R_N$$



$$I_s = \frac{R_N}{R_N + R_L} \times I_N$$

$$= \frac{4.12}{4.12 + 8} \times 1.714$$

$$= \underline{\underline{0.583 \text{ A}}}$$

~~53~~

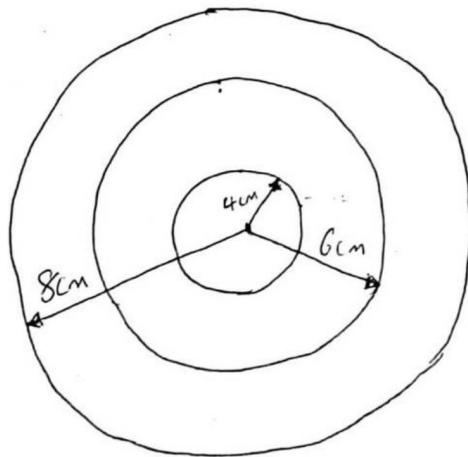
4(A)

i) mmf $\rightarrow$ 2013/2014 SESSION
Q 5A (iii)

ii) Magnetic flux Density

$\rightarrow$ 2015/2016
Q 16(i)

4(B)



54

Important NOTES: 4(6)

- 1) All Points inside a conducting Sphere are at the same potential as its Surface.
- 2) The field strength inside / within a conducting Sphere $E = 0$

At Point 2cm

$$E_2 = 0$$

$$V_2 = \frac{kq}{r} = \frac{9 \times 10^9 \times 8 \times 10^{-12}}{0.04} = 1.8 \text{ V}$$

At Point 5cm (within sphere 6cm)

$$V_5 = \frac{kq}{r_6} = \frac{9 \times 10^9 \times 8 \times 10^{-12}}{0.06} = 1.2 \text{ V}$$

~~SS~~

SS

$$E_5 = \frac{kq}{r_5^2} = \frac{9 \times 10^9 \times 8 \times 10^{-12}}{0.05^2}$$

$$= 28.8 \text{ N/C}$$

At Point 7 cm

$$V_7 = \frac{kq}{r_7} = \frac{9 \times 10^9 \times 4 \times 10^{-12}}{0.08}$$

$$= 0.45 \text{ V}$$

$$E_7 = \frac{kq}{r_7^2} = \frac{9 \times 10^9 \times 6 \times 10^{-12}}{(0.07)^2}$$

$$E_7 = 11 \text{ N/C}$$

~~56~~ 56

5(A)

- 1) Graphical method
- 2) Analytical method

5(B)

$$v(t) = 75 \sin(200\pi t - 0.25)$$

$$\omega = 200\pi$$

$$\phi = 0.25 \text{ rad}$$

$$B(i) \quad V_m = 75 \text{ V}$$

$$B(ii) \quad V_{p-p} = 2V_m = 2 \times 75 \\ = 150 \text{ V}$$

~~57~~

57.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFORUM
COLLEGE OF TECHNOLOGY
EXAMINATION: First Semester
COURSE CODE: EEE 211
TIME ALLOWED: 2 Hrs 30 mins



SESSION: 2015/2016
COURSE TITLE: Basic Electrical Engineering I
DEGREE: B.ENG

INSTRUCTION: Answer FOUR Questions with AT LEAST ONE Question from each section

SECTION A

1. (a) (i) State Coulomb's law of electrostatics. (1 ½ marks)
- (ii) State the inverse square law. (1 ½ marks)
- (iii) Calculate the electrostatic force of repulsion in free space when 10 pm is the distance apart between two particles. Both particles have an equal charge of 4×10^{-19} C and mass of 6.68×10^{-27} Kg. (2 marks)
- (b) (i) Define Magnetic Flux density. (1 ½ marks)
- (ii) With the aid of simple diagrams, outline three similarities and three dissimilarities between a magnetic circuit and an electric circuit. (3 ½ marks)
- (iii) A solenoid of length 20 cm has 500 turns and 5A current flowing through it. Calculate the magnetizing force and flux density. (1 mark)
- (c) (i) Explain briefly what you know about Inductors. (2 marks)
- (ii) Mention two applications of an inductor. (1 mark)
- (iii) A circular magnetic core with relative permeability of 2500 has a mean length of $5 \times 10^5 \mu\text{m}$ and a cross-sectional area of 2 cm^2 . Determine the current required in the coil of 2000 turns uniformly wound around the core to create a flux of $50 \mu\text{Wb}$ in iron. (3 ½ marks)

SECTION B

2. (a) (i) Explain what is meant by transient response. (2 ½ marks)
- (ii) For the circuit in figure 1, calculate the capacitor voltage 5 seconds after the switch is closed. (7 marks)
- (b) Calculate the mesh currents in the circuit in figure 2, and hence find the branch currents I_1 , I_2 and I_3 . (8 marks)

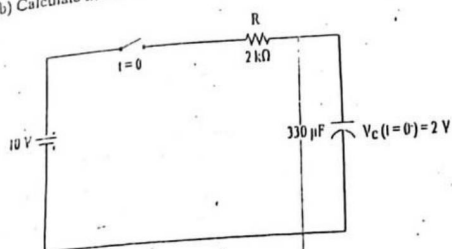


Figure 1

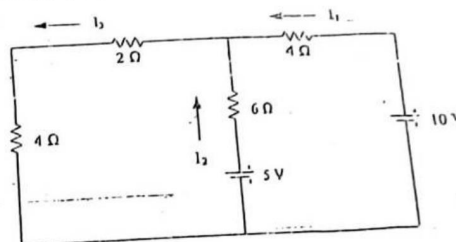


Figure 2

3. (a) (i) State Thevenin's theorem. (2 marks)
- (ii) Using Thevenin's theorem, calculate the current flowing through the 80Ω resistor in the circuit in figure 3. (8 ½ marks)

- (b) Applying Superposition theorem to the circuit in figure 4, calculate:
(i) I_2 (ii) V_{R2} (iii) the power consumed by resistor R_2 . (7 marks)

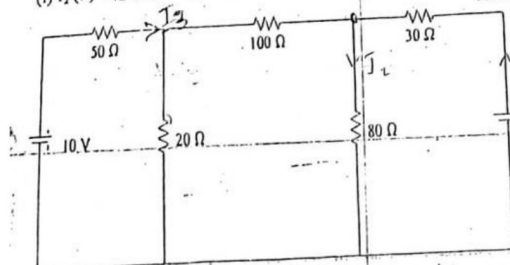


Figure 3

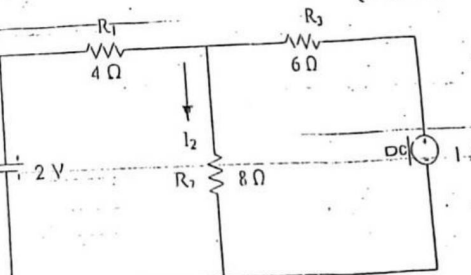


Figure 4

58

4. (a) (i) State the Maximum Power Transfer theorem.

(ii) Calculate the value of R which will absorb maximum power from the circuit in figure 5. Also, compute the value of the maximum power. (1 1/2 marks)

(b) Find the node voltages in the circuit in figure 6. (8 marks)

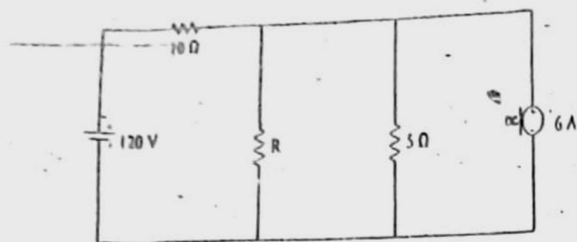


Figure 5

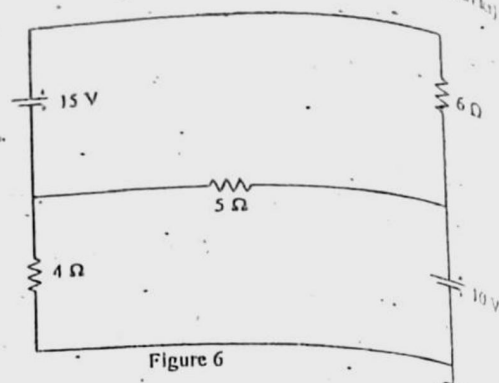


Figure 6

SECTION C

5. (a) Name three (3) types of alternating current other than sinusoidal (a.c) waveforms and state four (4) advantages of a.c. (2 marks)

(b) Use the analytical method to derive the equation for the root mean square (RMS) value of waveforms. Hence, if a sinusoidal waveform is defined by $V_p = 169.8 \sin(377t)$ volts, calculate the RMS voltage of the waveform, its frequency and the instantaneous value of the voltage after a time of 6 ms. (9 1/2 marks)

(c) State the conditions for resonance of an a.c waveform. A circuit has a resistance of 4.0Ω and inductance of 0.50 H and a variable capacitance in series is connected across a 100 V , 50 Hz supply. Calculate: (i) the value of the capacitance at resonance, (ii) the voltage across the inductance and the capacitance (iii) the Q factor of the circuit. (6 marks)

6. (a) Define the following terms associated with the a.c waveforms: (i) cycle (ii) period (iii) frequency. (3 marks)

(b) Derive the equation for the average value of waveforms. Calculate the average and effective values of the saw tooth waveform shown in Figure 7. The voltage completes the cycle by falling back to zero instantaneously after regular interval of time. (7 marks)

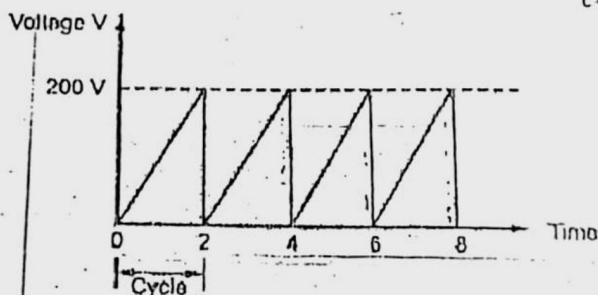


Figure 7

(c) Define reactive, active and apparent power. The voltage and current for a circuit having two elements in series are expressed as follows:

$$i(t) = 1.70 \sin(628t + \frac{\pi}{3})$$

$$v(t) = 8.5 \sin(628t + \frac{\pi}{2})$$

Determine: (a) Power factor (b) Frequency (c) Impedance (d) Resistance (e) Reactive element (f) Reactance and actual value of the reactive element (g) Active power. (7 1/2 marks)

2015/2016 SESSION

ANSWERS

1A
①

2012/2013 Q.1(a) 2nd Law

- ② The inverse Square Law states that the force of attraction btw two electrically charged particles is inversely proportional to the distance between the two charged particles.

~~11/11~~

60

1Aiii) Given;

$$d = 10 \text{ pm} = 10 \times 10^{-12} \text{ m} \\ = 10^{-11} \text{ m}$$

$$q_1 = q_2 = 4 \times 10^{-19} \text{ C}$$

$$F = \frac{k q_1 q_2}{d^2} ; \quad k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9$$

$$= \frac{9 \times 10^9 \times (4 \times 10^{-19})^2}{(10^{-11})^2}$$

$$= \frac{9 \times 10^9 \times 16 \times 10^{-38}}{10^{-22}}$$

$$= 144 \times 10^{-7} \text{ N}$$

$$= 14.4 \text{ MN}$$

~~50~~

61

(Aii) Given $L = 20\text{cm} = 0.2\text{m}$
 $N = 500 \text{ turns}$
 $I = 5\text{A}$

1B(i) This is defined as the amount of magnetic flux in an area taken perpendicular to the magnetic flux's direction.

1B(ii)

Similarities

- 1) Both Electric & Magnetic Circuit experiences an opposition (Resistance and Reluctance) to the flow of Current & flux respectively
- 2) The ^{total} Magnetomotive force like Electromotive force (E.m.f) of a ~~51~~ 62

Paper 4 of 5 — EEE 211 2016 2017

Level 200 • First Semester • PastQuestion

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COLLEGE OF TECHNOLOGY**

EXAMINATION: First Semester  SESSION: 2016/2017
COURSE CODE: EEE 211 COURSE TITLE: Basic Electrical Engineering I
TIME ALLOWED: 2Hrs 30mins DEGREE: B.ENG
INSTRUCTION: Answer FOUR Questions from both sections

SECTION A: Answer TWO Questions from this section

1. (a) State Kirchhoff's current law and Kirchhoff's voltage law. (3 marks)

(b) (i) Obtain the node voltages in the circuit of figure 1 using nodal analysis. (7 marks)

(ii) Using Superposition theorem, calculate the current flowing through the $4\ \Omega$ resistor in figure 2. (7 ½ marks)

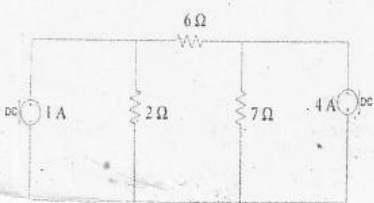


Figure 1

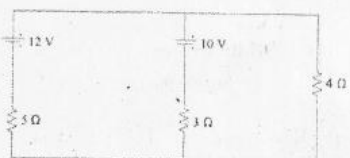


Figure 2

2. (a) Explain the term transient response. (2 marks)

(b) (i) Calculate the mesh currents in the circuit of figure 3. (7 marks)

(ii) In the circuit of figure 4, find the value of R_L necessary for maximum power transfer, and hence calculate the maximum power that can be delivered to R_L . (8 ½ marks)

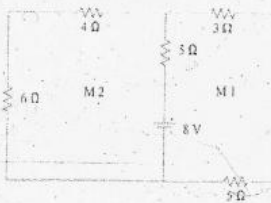


Figure 3



Figure 4

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3. (a) State Thevenin theorem and Norton theorem. (5 marks)

(b) (i) Obtain the Thevenin equivalent circuit and calculate the current flowing through the $8\ \Omega$ resistor in figure 5. (9 marks)

(ii) Draw the Norton equivalent circuit of the circuit obtained in Question 3b(i) and verify the value of the current flowing through the $8\ \Omega$ resistor. (3 ½ marks)

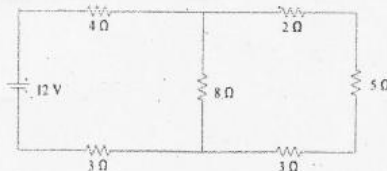


Figure 5

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4. (a) Define briefly the following (i) Magneto motive force (mmf) (ii) Magnetic flux density.

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(b) An alternating voltage is given by $v = 75 \sin(200\pi t - 0.25)$ volts. Find (i) the amplitude, (ii) the peak-to-peak value, (iii) the r.m.s. value, (iv) the periodic time, (v) the frequency, and (vi) the phase angle (in degrees and minutes) relative to $75 \sin 200\pi t$. (17 ½ marks)

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(ii) A series circuit of resistance $60\ \Omega$ and inductance 75 mH is connected to a 110 V, 60 Hz supply. Calculate the power dissipated. (17 ½ marks)

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Paper 5 of 5 — EEE 211 PAST QUESTION

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY

EXAMINATION: First Semester **SESSION: 2016/2017**
COURSE CODE: EEE 211 **COURSE TITLE: Basic Electrical Engineering I**
TIME ALLOWED: 2Hrs 30mins **DEGREE: B.ENG**

INSTRUCTION: Answer FOUR Questions from both sections

SECTION A: Answer TWO Questions from this section

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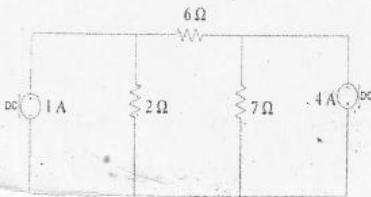


Figure 1

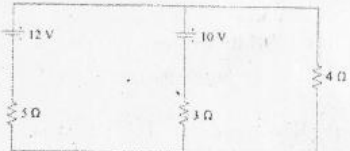


Figure 2

2. (a) Explain the term transient response. (2 marks)
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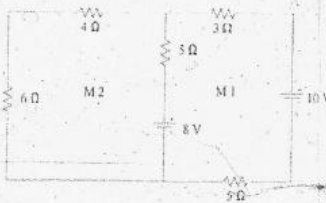


Figure 3

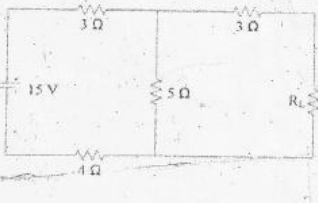


Figure 4

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3. (a) State Thevenin theorem and Norton theorem. (5 marks)

(b) (i) Obtain the Thevenin equivalent circuit and calculate the current flowing through the $8\ \Omega$ resistor in figure 5. (9 marks)

(ii) Draw the Norton equivalent circuit of the circuit obtained in Question 3b(i) and verify the value of the current flowing through the $8\ \Omega$ resistor. (3 ½ marks)

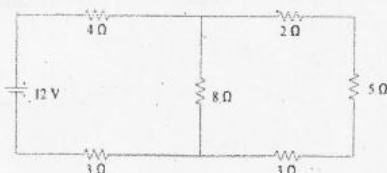


Figure 5

SECTION B: Answer TWO Questions from this section

4. (a) Define briefly the following (i) Magneto motive force (mmf) (ii) Magnetic flux density.

(b) Three concentric spheres of radii 4, 6, and 8 cm have charges of +8, -6 and +4 μC respectively. What are the potentials and field strengths at points 2 cm, 5 cm, 7 cm and 10 cm from the centre? (17 ½ marks)

5. (a) List the methods used in analyzing alternating current (ac).

(b) An alternating voltage is given by $v = 75 \sin(200\pi t - 0.25)$ volts. Find (i) the amplitude, (ii) the peak-to-peak value, (iii) the r.m.s. value, (iv) the periodic time, (v) the frequency, and (vi) the phase angle (in degrees and minutes) relative to $75 \sin 200\pi t$. (17 ½ marks)

6. (a) Define the following: (i) Apparent power (ii) Power factor.

(b) (i) A load takes 90 kW at a power factor of 0.5 lagging. Calculate the apparent power and the reactive power.

(ii) A series circuit of resistance $60\ \Omega$ and inductance 75 mH is connected to a 110 V, 60 Hz supply. Calculate the power dissipated. (17 ½ marks)

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